

**RESILIENCE WITHOUT DISCRIMINATION: RECONCEPTUALIZING SUBSIDY
DISCIPLINES IN THE ERA OF INDUSTRIAL POLICY****Ru Ding*****Not for circulation, Draft - 6 May 2026****ABSTRACT**

Supply chain resilience has become a core objective of economic security and industrial policy. Governments increasingly deploy subsidies to induce investment in slack and flexibility, improve information, and speed adjustment to shocks. Many of these programs are also politically contested because they appear to privilege domestic production or selected foreign partners. This Article argues that resilience policy does not require discriminatory subsidization, and that WTO law can accommodate resilience aims while maintaining non-discrimination as the baseline. Drawing on economic accounts of resilience, it shows that resilience is not always underprovided and that public support can also induce overinvestment. It argues that the case for intervention is product, market, and risk specific, and cautions against one-size-fits-all resilience subsidies. Building on this framework, the Article maps recurring design patterns in recent practice of U.S., EU, China, Japan, South Korea, etc. It contends that the greatest legal vulnerability arises when eligibility or benefit levels turn on the origin of content, especially through local or designated foreign content conditions. By contrast, many forms of conditionality commonly used in resilience programs, including requirements tied to the nationality of the producer or to territorial presence, often fall short of a breach once the risk of disruption is calibrated with the measures taken. The Article therefore shifts the focus from broad justifications under general exceptions to the institutional tools that can discipline resilience subsidies without converting discrimination into a default policy instrument. It proposes a path built on transparency and domestic review, proportionate countervailing responses, and gradual rule making that links support to measurable resilience outcomes. The result is a framework that preserves policy space while keeping discrimination exceptional rather than routine.

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I. INTRODUCTION

Supply chain resilience has become a major theme in the reorganization of the global economic order. During the Biden administration, resilience was framed as an economic security priority. Executive Order 14017 declared that the U.S. needs “resilient, diverse, and secure supply chains” to ensure “economic prosperity and national security.”¹ The global impulse toward resilience did not wane as U.S. administrations changed. At the Sixteenth United Nations Conference on Trade and Development (UNCTAD) in Geneva in October 2025, UNCTAD Secretary General Rebeca Grynspan captured what disruption now means in practice. “A ship that once crossed the Red Sea in days now sails for weeks around the Cape of Good Hope.”² Since the early phase of the COVID-19 pandemic, governments have adopted trade and industrial measures across sectors including medical goods,³ semiconductors,⁴ and critical minerals.⁵

Building resilience jumps out the binary of rewards and risks underpinning international economics law.⁶ Supply chain resilience does not only connote what are the measures to be taken when risks are imminent, but also emphasizes the preparedness and adaptiveness for the longer term when the risks of disruption actually materialize. This

¹ Exec. Order No. 14017, 86 Fed. Reg. 11849 (Feb. 24, 2021).

² *Global Supply Chains Under Strain – Ministers Call for Just and Resilient Transitions*, U.N. CONF. TRADE & DEV. (Oct. 22, 2025), <https://unctad.org/news/global-supply-chains-under-strain-ministers-call-just-and-resilient-transitions>.

³ Philip Blenkinsop, *EU Frees Up Export of Coronavirus Protection Equipment*, REUTERS (May 26, 2020), <https://www.reuters.com/article/business/healthcare-pharmaceuticals/eu-frees-up-export-of-coronavirus-protection-equipment-idUSKBN2321WL/>; Vishwadha Chander, *U.S. Awards New \$628 Million Contract to Boost Output of Potential COVID-19 Vaccine*, REUTERS (June 1, 2020), <https://www.reuters.com/article/world/us-politics/us-awards-new-628-million-contract-to-boost-output-of-potential-covid-19-vacc-idUSKBN2382DN/>;

⁴ Stephen Nellis, Karen Freifeld & Alexandra Alper, *U.S. Aims to Hobble China’s Chip Industry with Sweeping New Export Rules*, REUTERS (Oct. 10, 2022), <https://www.reuters.com/technology/us-aims-hobble-chinas-chip-industry-with-sweeping-new-export-rules-2022-10-07/>; David Shepardson, *U.S. Finalizes More Than \$6.1 Bln Chips Subsidy for Micron Technology*, Reuters (Dec. 10, 2024), <https://www.reuters.com/technology/us-finalizes-more-than-61-bln-chips-subsidy-micron-technology-2024-12-10/>.

⁵ Julia Payne, *EU Efforts to Diversify Critical Raw Material Imports Fail So Far, Auditors Say*, REUTERS (Feb. 2, 2026), <https://www.reuters.com/sustainability/climate-energy/eu-efforts-diversify-critical-raw-material-imports-fail-so-far-auditors-say-2026-02-02/>; Colleen Howe & Liz Lee, *China Imposes Export Controls on 20 Japanese Entities to Curb “Remilitarisation”*, REUTERS (Feb. 24, 2026), <https://www.reuters.com/world/asia-pacific/china-adds-20-japanese-entities-export-control-list-2026-02-24/>.

⁶ See generally, Anthea Roberts, *Risk, Reward, and Resilience Framework: Integrative Policy Making in a Complex World*, 26 J. INT’L ECON. L. 233 (2023).

framing has an institutional implication. If a government seeks to incentivize or intervene in capability formation, subsidies are a preferred instrument because they can accelerate long term investments in capacity, redundancy, information, and adjustment.⁷

At the same time, resilience measures come with real costs. From a macroeconomic perspective, relocation, whether domestic or toward designated destinations, can be costly, and the global economy can bear the welfare losses. The Organization of Economic Cooperation and Development (OECD) issued its first Supply Chain Resilience Review. In this review, the OECD found that,

“[s]ome policies aiming to make value chains more domestic (relocalised) could be costly and may not necessarily offer more stability in the face of shocks []. Compared to an interconnected regime, relocalisation [], could decrease global trade by more than 18% and global real GDP by more than 5%, with individual countries losing between 1.1 and 12.2% of GDP depending on the extent and nature of their GVC integration. In addition, the relocalised regime was no more resilient to shocks – in terms of the stability of GDP, production and consumption (i.e. deviations of these variables from the baseline levels in reaction to shocks) – than the interconnected regime. In fact, for more than half of the economies, the stability of GDP decreased in the localized regime. This illustrates that openness and the geographical diversification of input sources and output destinations in GVCs can offer important options for adjusting to disruptions and mitigating exposure to shocks from a greater number of sources.”⁸

From a microeconomic perspective, firms can also misallocate resilience investments. Some firms may underinvest where resilience generates external benefits for downstream users and the wider production network. Other firms may overinvest where scarcity pricing and business stealing make private returns exceed social value.⁹ These dynamics mean that

⁷ Chad P. Brown, *Modern Industrial Policy and the World Trade Organization*, 16 ANN. REV. ECON. 243, 259-61 (2024); Réka Juhász, Nathan J. Lane & Dani Rodrik, *The New Economics of Industrial Policy* 4 (NBER Working Paper No. 31538, 2023), https://www.nber.org/system/files/working_papers/w31538/w31538.pdf (explaining that industrial policy operates by creating incentives for firms, innovators, and investors to undertake desired investments, and describing subsidies for investment and R&D as the most obvious instruments).

⁸ Org. Econ. Coop. Dev., *OECD Supply Chain Resilience Review: Navigating Risks* 16 (June 2, 2025), https://www.oecd.org/content/dam/oecd/en/publications/reports/2025/06/oecd-supply-chain-resilience-review_9930d256/94e3a8ea-en.pdf.

⁹ For discussion on the two externalities, see Kiminori Matsuyama & Philip Ushchev, *When Does Procompetitive Entry Imply Excessive Entry?* (CEPR Discussion Paper No.

subsidies can be welfare enhancing in some settings and wasteful in others. The central question becomes how to appraise these measures and how to discipline their design.

Stripping away the policy justification of resilience-building, two preliminary legal observations shall be made. First, resilience subsidies must be evaluated under the World Trade Organization (WTO) Agreement on Subsidies and Countervailing Measures (SCM Agreement) because they may distort competition and harm the interests of other Members through adverse effects.¹⁰ Second, and less discussed in the subsidy literature, many resilience subsidies have a discriminatory facet. States may favor inputs from designated reliable sources, subsidize sourcing in selected jurisdictions, or limit eligibility to firms established in, or linked to, particular countries. These features are to be scrutinized under the General Agreement on Tariffs and Trade (GATT) non-discrimination obligations under Articles I and III.¹¹ A further question then emerges in hard cases. If a resilience subsidy is inconsistent with core obligations, can it be justified under the GATT exceptions. Recent litigation has placed these issues on the WTO docket, including the Panel report circulated in January 2026 in *U.S.–IRA* concerning certain U.S.’s tax credits for clean electronic vehicle and clean energy programs.¹²

In this article, we adopt a broad understanding of subsidies, namely any measure, including any form of income or price support, within the meaning of GATT Article XVI, including measures that operate directly or indirectly to increase exports or reduce imports.¹³ First, we provide a descriptive account of contemporary resilience-oriented subsidization and develops an analytical typology that organizes state practice by the object of selectivity and the location of discrimination along the supply chain. Second, we advance a doctrinal claim that the existing WTO framework generally permits a substantial share of these measures. It shows that even though the exceptions under the GATT might not be workable, the SCM Agreement and the GATT grants significant policy space for states to subsidize supply chain resilience (and states do observe the limitations in practical sense). We argue that as the baseline, subsidies contingent upon designated inputs or indirect producer subsidies conferring benefits on consumers are violating Articles I or III of the

14991, 2021), <https://cepr.org/publications/dp14991>.

¹⁰ Agreement on Subsidies and Countervailing Measures, Apr. 15, 1994, Marrakesh Agreement Establishing the World Trade Organization, Annex 1A, 1869 U.N.T.S. 14 [hereinafter SCM Agreement].

¹¹ General Agreement on Tariffs and Trade, Oct. 30, 1947, 61 Stat. A-11, 55 U.N.T.S. 194 [hereinafter GATT].

¹² Panel Report, *United States–Certain Tax Credits Under the Inflation Reduction Act*, WTO Doc. WT/DS623/R (circulated Jan. 30, 2026) [hereinafter U.S.–IRA Panel Report].

¹³ GATT art. XVI.

GATT since their market effects are direct and harmful. Third, we develop a normative argument that the role of non-discrimination under the WTO law, including the subsidy discipline, are pertinent and important despite political disenchantment. We devise domestic self-discipline, regional cooperation and potentially countervailing measures to control the unpermitted discrimination in supply chain resilience subsidies.

The Article proceeds as follows. Part II develops the economic case for resilience and identifies the conditions under which market incentives diverge from social welfare. Part III maps contemporary state practice and presents the typology of strategic selectivity. Part IV evaluates legitimacy under the SCM Agreement and the GATT, including the scope of relevant exceptions. Part V proposes a way forward for disciplining resilience subsidies while preserving workable policy space. Part VI concludes.

II. THE ECONOMY CASE OF SUPPLY-CHAIN RESILIENCE

This Part serves two functions. First, it defines supply chain resilience and identifies what it requires. It does so by distinguishing the term's usage in international political discourse, including characterizations that call for reform of particular regimes of international economic law, from the definitions commonly adopted in the logistics and management literature. Second, it examines both firm-led resilience initiatives and state-led public policies intended to shape private firms' resilience choices. The objective is to clarify when resilience initiatives can be left to market forces and when government intervention is warranted, either to encourage socially necessary resilience or to discourage inefficient measures.

A. What is Supply Chain Resilience and the Current Research

Politically, supply chain resilience had been treated as an economic security objective. In the U.S., former President Biden's Executive Order 14017 framed "resilient, diverse, and secure supply chains" as necessary for economic prosperity and national security and launched the White House's 100-day supply chain reviews program to scrutinize domestic production capacity.¹⁴ Following this framing of supply chain resilience as a national and economic security priority, other heads of state and government echoed similar themes. Singapore's Prime Minister Lee Hsien Loong warned that the turn toward "resilience and self-sufficiency" must be carefully calibrated to avoid deeper bifurcation of trade and

¹⁴ Exec. Order No. 14017, 86 Fed. Reg. 11849 (Feb. 24, 2021).

technology, while still strengthening supply chains against shocks.¹⁵ India's Prime Minister Narendra Modi, speaking at the Global Summit on Supply Chain Resilience, called for supply to come from "trusted sources," for greater transparency, and for time-bound coordination to prevent tit-for-tat disruptions.¹⁶ South Korea's former President Yoon Suk Yeol, in his Davos special address, described strengthening global supply-chain resilience as "the most urgent task" and tied it to international solidarity and cooperative rule-based action.¹⁷ At multilateral forums, WTO MC12's Ministerial Declaration on the WTO Response to COVID-19 and Preparedness for Future Pandemics linked preparedness to diversification and the reduction of barriers for essential goods.¹⁸

States start to promulgate legislations to enhance supply chain resilience. Domestically, Japan's Economic Security Promotion Act establishes mechanisms aimed at ensuring the stable supply of critical products and other security-related economic measures.¹⁹ South Korea's Framework Act on Supply Chain Stabilization Support for Economic Security similarly targets the prevention and management of supply-chain risks through state-led stabilization and response measures.²⁰ In the EU, the Critical Raw Materials Act frames "supply risk preparedness and mitigation" through monitoring, stress-testing, and coordinated strategic measures (including stock-related and corporate risk-preparedness components).²¹ At the international level, the IPEF Agreement Relating to

¹⁵ *Intervention by PM Lee Hsien Loong at the Global Summit on Supply Chain Resilience*, PRIME MINISTER OFF. SING. (Oct. 31, 2021), <https://www.pmo.gov.sg/newsroom/pm-lee-at-the-global-summit-on-supply-chain-resilience/>.

¹⁶ *Prime Minister's address at Global Summit on Supply Chain Resilience*, MINISTRY EXTERNAL AFF. INDIA (Oct. 31, 2021), <https://www.mea.gov.in/Speeches-Statements.htm?dtl%2F34458%2Fprime+ministers+address+at+global+summit+on+supply+chain+resilience=>.

¹⁷ *Davos 2023: Special Address by Yoon Suk Yeol, President of the Republic of Korea*, WORLD ECON. F. (Jan. 19, 2023), <https://www.weforum.org/stories/2023/01/davos-2023-special-address-by-yoon-suk-yeol-president-of-the-republic-of-korea/>.

¹⁸ World Trade Organization, Ministerial Declaration on the WTO Response to the COVID-19 Pandemic and Preparedness for Future Pandemics, WTO Doc. WT/MIN (22)/31 (2022).

¹⁹ Keizai shisaku o ittaiteki ni kōzu koto ni yoru anzen hoshō no kakuho no suishin ni kansuru hōritsu [Keizai Anzen Hoshō Suishinhō] [Act on the Promotion of Ensuring National Security Through Integrated Implementation of Economic Measures], Law No. 43 of 2022, art. 1 (Japan).

²⁰ Gyeongje anbo-reul wihan gonggeummang anjeonghwa jiwon gibonbeop [Framework Act on Supply Chain Stabilization Support for Economic Security], Act No. 19828, Dec. 26, 2023, art. 1 (S. Kor.).

²¹ Regulation (EU) 2024/1252 of the European Parliament and of the Council of 11 Apr. 2024 Establishing a Framework for Ensuring a Secure and Sustainable Supply of Critical Raw Materials and Amending Regulations (EU) No 168/2013, (EU) 2018/858, (EU) 2018/1724 & (EU) 2019/1020, O.J. (L) 2024/1252 (May 3, 2024), arts. 20(1), (3), 22-23, 24.

Supply Chain Resilience provides a treaty-form example that creates institutional mechanisms and cooperation commitments aimed at preventing, mitigating, and responding to supply-chain disruptions among parties.²² The shift from mere political statements to international and domestic norms is conceptually important. Resilience becomes not only a policy aspiration, but a legal basis for designating critical sectors, requiring planning and monitoring, and structuring inter-governmental coordination that can discipline both public and private conduct.

Scholarly accounts trace the term “supply chain resilience” to earlier work on “supply chain vulnerability.”²³ In 2001, the Center for Logistics and Supply Chain Management examined supply chain vulnerability during the 2000 fuel protests and the outbreak of foot and mouth disease in the UK. That study offered a working definition of supply chain vulnerability as “an exposure to serious disturbance, arising from risks within the supply chain as well as risks external to the supply chain.” It also found that the U.K. public sector had well established practices for addressing vulnerability, while many industries were less attentive to the relevant risks. The study’s conclusions then used “supply chain resilience” as the organizing term for subsequent programs.²⁴ Earlier clear-cut definition adopted “resilience” as “the ability of a system to return to its original state or move to a new, more desirable state after being disturbed.” This suggested the notion of flexibility, and the adaptability of supply chain to newer change.²⁵ Others use it in a more general manner to capture “the ability [of a supply chain] to bounce back from a disruption,”²⁶ highlighting the speed of returning to normal performance levels following low-likelihood, high-impact disruptions.

Over time, supply chain resilience has come to encompass more than responsiveness and risk management. It also emphasizes adaptability and a forward looking orientation. Joseph Fiksel incorporated the corporation’s capacity to survive, adapt, and grow amid

²² Indo-Pacific Economic Framework for Prosperity, Nov. 14, 2023, 24 Stat. 224. For an evaluation of its relationship with and legitimacy under the WTO law, see Christian Delev, *Economic Resilience Under the Indo-Pacific Economic Framework Supply Chain Agreement: Treading between Geoeconomic Ambitions and World Trade Organization Compatibility?*, 24 WORLD TRADE REV. 645 (2025).

²³ Timothy J. Pettit, Joseph Fiksel & Keely L. Croxton, *Ensuring Supply Chain Resilience: Development of a Conceptual Framework*, 31 J. BUS. LOGISTICS 1, 4 (2011).

²⁴ See *Creating Resilient Supply Chains: A Practical Guide*, Centre for Logistics and Supply Chain Management, CRANFIELD SCH. MGMT. (2003), <http://hdl.handle.net/1826/4374>.

²⁵ Martin Christopher & Helen Peck, *Building the Resilient Supply Chain*, 15 INT’L J. LOGISTICS MGMT. 1, 2 (2004).

²⁶ Yossi Sheffi & James B. Rice Jr., *A Supply Chain View of the Resilient Enterprise*, 47 MIT SLOAN MGMT. REV. 41, 41 (2005).

turbulent change.²⁷ Timothy J. Pettit, Keely L. Croxton, and Joseph Fiksel conceptualized “supply chain resilience” through the binary of vulnerability and capacity. They maintained two postulates: forces of change create supply chain vulnerabilities; management controls create supply chain capacities. They argued that resilience tends to increase as capacities rise and vulnerabilities fall, yet it requires balance, since excess on either side can produce greater risk exposure or reduced profitability.²⁸ The literature also operationalizes supply chain resilience through a portfolio of elements. In the *ex ante*, or readiness, phase, firms anticipate disruptions through information sharing, human resource management, strategic inventory management, and redundancy across multiple suppliers.²⁹ In post disruption phases, firms seek agility, maintain collaboration, strengthen future flexibility, and build redundancy to absorb sudden changes.³⁰

International law scholars have paid increasing attention to supply chain resilience, particularly in light of the COVID-19 disruptions and the Russia-Ukraine conflict. At a general level, Timothy Meyer argued that trade liberalization may do little to strengthen supply chain resilience, and that the existing trade architecture can constrain preventive supply chain regulation through non-discrimination obligations, without a workable carve-out in the exceptions.³¹ Building from a normative commitment to free trade, Xuanli Chen and Wei Shen contended that WTO litigation can be used to challenge discriminatory supply chain policies.³² In more specific analyses, Andrew D. Mitchell used the example

²⁷ Joseph Fiksel, *Sustainability and Resilience: Toward a Systems Approach*, 2, SUSTAINABILITY SCI. PRAC. & POL’Y 14 (2006).

²⁸ Pettit, Fiksel & Croxton, *supra* note 23, at 6-7; Timothy J. Pettit, Keely L. Croxton & Joseph Fiksel, *Ensuring Supply Chain Resilience: Development and Implementation of an Assessment Tool*, 34 J. BUS. LOGISTICS 46 (2013).

²⁹ Uta Jüttner & Sarah Maklan, *Supply Chain Resilience in the Global Financial Crisis: An Empirical Study*, 16 SUPPLY CHAIN MGMT.: AN INT’L J. 246 (2011); Daniel Kern et al., *Supply Risk Management: Model Development and Empirical Analysis*, 42 INT’L J. PHYSICAL DISTRIB. & LOGISTICS MGMT. 60 (2012); Paul R. Kleindorfer & Germaine H. Saad, *Managing Disruption Risks in Supply Chains*, 14 PROD. & OPERATIONS MGMT. 53 (2005); Christopher W. Craighead et al., *The Severity of Supply Chain Disruptions: Design Characteristics and Mitigation Capabilities*, 38 DECISION SCI. 131 (2007).

³⁰ Andreas Wieland & Carl Marcus Wallenburg, *The Influence of Relational Competencies on Supply Chain Resilience: A Relational View*, 43 INT’L J. PHYSICAL DISTRIB. & LOGISTICS MGMT. 300 (2013); Christoph Bode et al., *Understanding Responses to Supply Chain Disruptions: Insights from Information Processing and Resource Dependence Perspectives*, 54 ACAD. MGMT. J. 833 (2011); George A. Zsidisin & Stephan M. Wagner, *Do Perceptions Become Reality? The Moderating Role of Supply Chain Resiliency on Disruption Occurrence*, 31 J. BUS. LOGISTICS 1 (2010).

³¹ Timothy Meyer, *Trade Law and Supply Chain Regulation in a Post-COVID-19 World*, 114 AM. J. INT’L L. 637 (2020).

³² Xuanli Chen & Wei Shen, *Disputing Supply Chain Security Legislation in the Context of WTO Rules: Litigability, Legitimacy, and Exceptionality*, 16 J. INT’L DISP. SETTLEMENT

of onshoring pharmaceuticals as a resilience oriented response and highlights the public health rationale for such measures.³³ His later work considered how these measures might be designed to remain consistent with WTO disciplines.³⁴ Also in the context of medical products, Chad P. Brown examined the economic case for vaccine production and its international supply chains, and argued for plural initiatives that sustain vaccine subsidies while constraining export restrictions.³⁵

In the field of critical minerals, Weihuan Zhou, Victor Crochet, and Haoxue Wang challenged the simplified narratives of “de-risking” by examining China’s critical minerals strategy and emphasizing the complexity of state tools deployed in pursuit of supply security.³⁶ Oliver Hailes, through the lens of lithium governance, situated resilience within broader trade, investment, and distributive justice concerns, arguing that supply-chain regulation must be assessed not only in efficiency terms but also in light of equity and sustainability.³⁷ Victor Crochet and Weihuan Zhou similarly analyzed the EU’s critical minerals strategy as a hybrid of internal regulatory discipline and external partnership diplomacy, raising questions about industrial policy and trade compatibility.³⁸ Thomas Volland and Shona Daly examined the EU Conflict Minerals Regulation as a due-diligence-based governance model that addresses upstream risks through corporate obligations rather than conventional trade barriers.³⁹ Reinhard Biedermann argued that global raw-materials governance is shifting from WTO-centered trade liberalism toward fragmented post-liberal regime inflected by sovereignty and sustainability.⁴⁰

Against this background, “supply chain resilience” in this paper refers to a balanced configuration of vulnerability and capacity in production networks that enables firms to

1 (2025).

³³ Andrew D. Mitchell, *The Geography of Health: Onshoring Pharmaceutical Manufacturing to Address Supply Chain Challenges*, 23 WORLD TRADE REV. 519 (2024).

³⁴ Andrew D. Mitchell, *Home Remedies: Flexibilities to Onshore Pharmaceutical Manufacturing under WTO Rules*, 28 J. INT’L ECON. L. 23 (2025).

³⁵ Chad P. Brown, *The WTO and Vaccine Supply Chain Resilience During a Pandemic*, 26 J. INT’L ECON. L. 343 (2023).

³⁶ Weihuan Zhou, Victor Crochet & Haoxue Wang, *Demystifying China’s Critical Minerals Strategies: Rethinking ‘de-Risking’ Supply Chains*, 24 WORLD TRADE REV. 257 (2025).

³⁷ Oliver Hailes, *Lithium in International Law: Trade, Investment, and the Pursuit of Supply Chain Justice*, 25 J. INT’L ECON. L. 148 (2022).

³⁸ Victor Crochet & Weihuan Zhou, *Critical Insecurities? The European Union’s Strategy for a Stable Supply of Minerals*, 27 J. INT’L ECON. L. 147 (2024).

³⁹ Thomas Volland & Shona Daly, *The EU Regulation on Conflict Minerals: The Way out of a Vicious Cycle?*, 52 J. WORLD TRADE 37 (2018).

⁴⁰ Reinhard Biedermann, *From Trade Liberalism to Resource Sovereignty: Paradigm Change in Global Raw Materials Governance*, 60 J. WORLD TRADE 23 (2026).

maintain, or to recover rapidly to, economically meaningful levels of output and availability under low probability. Under this account, resilience increases as vulnerabilities are reduced and relevant capacities are strengthened, yet the target is an economically necessary level of capacity that avoids inefficient redundancy. The political and legal materials discussed above indicate that resilience has also become a governmental objective that integrates economic security concerns and trade liberalization goals. These initiatives tend to emphasize ex ante preparedness more than ex post adaptation. This emphasis frames the core normative question for the subsidy analysis. The case for governments to use subsidies to strengthen supply chain resilience depends on whether market forces are sufficient to induce the relevant investments and on the cross-border spillover effects of those policies on other states.⁴¹

B. Private Firm's Strategic Initiative to Supply Chain Resilience

There are insurmountable pieces of literature addressing firms' actions and incentives to enhance supply chain resilience in management and logistic literature.⁴² However, only scarcely the problem of whether private firms' strategic supply chain resilience initiatives are efficient or inefficient are covered. Gene M. Grossman, Elhanan Helpman and Alejandro Sabal developed a general-equilibrium model of multi-tier vertical supply chains to study whether private firms invest efficiently in supply-chain resilience.⁴³ The model is based on the sequential production of final products across multiple tiers, during which bilateral Nash-in-Nash bargaining between upstream and downstream firms reappears, while producers engage in endogenous formation of supplier networks (diversification), and investments in firms' own resilience. The authors emphasize two levels of distortion. First, investments in resilience and network formation generate positive vertical externalities, since when an upstream firm survives a disruption, downstream firms can continue production. Upstream firms hence do not internalize these benefits, and the social incentives is higher than the firm-level incentives. Firms tend to underinvest. Second, bargaining markups distort firms' private returns to resilience, since upstream firms earn scarcity rents that exceed the real social value of continued production. In this case, midstream firms tend to overinvest in resilience.⁴⁴ Whether a firm overinvest or

⁴¹ Gene M. Grossman & Alan O. Sykes, *Industrial Policy and Subsidies: Assessment of Current Rules and Possible Reforms*, 59 J. WORLD TRADE 531, 540-42 (2025).

⁴² See William Ho et al., *Supply Chain Risk Management: A Literature Review*, 53 INT. J. PROD. RSCH. 5031 (2015).

⁴³ Gene M Grossman, Elhanan Helpman & Alejandro Sabal, *Optimal Resilience in Multitier Supply Chains*, 139 Q.J. ECON. 2377 (2024).

⁴⁴ *Id.*, at 2414.

underinvest depends on its place on the supply chain.⁴⁵ The same holding is also argued in another work by Grossman, Helpman and Lhuillier, though emphasizing the business-stealing externalities incentivizing firms in a “resilience race” to be the one that supply during a shock and capture high post-crisis profits.⁴⁶

Agostino Capponi, Chuan Du and Joseph E. Stiglitz agreed with the first part understanding of Grossman, Helpman and Sabal, arguing that upstream producers tend to underinvest in supply chain resilience. Different from Grossman, Helpman and Sabal, they model resilience as investment in non-scalable production capacity that must be installed before shocks are realized, and then firms adjust only “scalable inputs” after the shock.⁴⁷ This is what makes *ex ante* resilience costly and potentially inefficient if overdone. Two structural frictions create market failure. A finite number of intermediate suppliers with imperfectly substitutable inputs have upstream market power to sustain markups when the supply is insufficient. Additionally, firms cannot insure fully for supply chain disruption, hence they rely on re-order contracts before shocks and spot markets after shocks. Through incomplete contracting theory and scarcity pricing in spot markets, Capponi, Du and Stiglitz concluded that adding capacity is socially valuable because it prevents shortages and price spikes, but it is privately less attractive because more capacity lowers the spot price in disruptive times, and much of the benefit goes to buyers rather than the capacity owner.⁴⁸

Daron Acemoglu and Alireza Tahbaz-Salehi provided a macroeconomic foundation for why privately chosen supply-chain structures can be inefficiently fragile even when firms behave rationally.⁴⁹ Rather than focusing on firm-level resilience or capacity expansion, they emphasize the formation and severance of relationship-specific production links in networks where inputs are not perfectly substitutable and surplus is divided through bargaining.⁵⁰ In their framework, supply chains are built through costly customized connections and firms retain the option to cut links when profitability deteriorates after

⁴⁵ *Id.*, at 2382.

⁴⁶ Gene M. Grossman, Elhanan Helpman & Hugo Lhuillier, *Supply Chain Resilience: Should Policy Promote International Diversification or Reshoring?*, 131 J. POL. ECON. 3462 (2023).

⁴⁷ Agostino Capponi, Chuan Du & Joseph E. Stiglitz, *Are Supply Networks Efficiently Resilient?* 3-4 (NBER Working Paper Series No. 32221), https://www.nber.org/system/files/working_papers/w32221/w32221.pdf.

⁴⁸ *Id.*, at 24-25.

⁴⁹ Daron Acemoglu & Alireza Tahbaz-Salehi, *The Macroeconomics of Supply Chain Disruptions*, 92 REV. ECON. STUD. 656 (2025).

⁵⁰ *Id.*, at 25.

disruptions.⁵¹ The central inefficiency arises because each bilateral link creates positive spillovers on the productivity and feasibility of production for other firms in the network, yet the parties to the link do not internalize these system-wide benefits when deciding whether to maintain, drop, or reconfigure relationships.⁵² As a result, privately optimal responses to disruptions can trigger cascading failures and discontinuous declines in aggregate output that are larger than the initial shock. The implication is that even with fully rational private behavior, the market equilibrium may feature under-maintenance and under-formation of resilient network connections relative to the social optimum, not because firms misperceive risks, but because network externalities and bargaining prevent them from capturing the broader macroeconomic gains of a more robust production web.⁵³

Several legal works taken upon cooperation's initiatives to resilience carefully.⁵⁴ Drawing upon Grossman, Helpman and Lhuillier, Doni Bloomfield and Jeff Gordon used the opposing resilience externality and business-stealing externality to evaluate firms' incentives.⁵⁵ For resilience externality, they similarly argued that resilience is socially valuable in production networks and firms under-appropriate the gains from being reliable because much of resilience shows up as consumer surplus and as spillovers to diffuse third parties. In addition, they added that resilience failures are under-punished, in part because market power and crisis markups soften the private downside of disruption, and *ex post* public interventions can create classic moral hazard. For business-stealing externality, however, Bloomfield and Gordon argued what looks like wasteful rivalry often produces large spillover benefits. Firms often cannot fully cash in on crisis scarcity. If firms expect the government to step in during emergencies, the extra payoff from being the lone firm shrinks. Additionally, many critical inputs are not interchangeable, so product variety has independent social value. Hence, Bloomfield and Gordon maintained that insufficient resilience is the more empirically plausible and socially costly error.⁵⁶

C. Governmental Intervene in Resilience Building

In supply chain literature, public policies are commonly viewed as the variant or the

⁵¹ *Id.*

⁵² *Id.*, at 4-5.

⁵³ *Id.*, at 23-25.

⁵⁴ Matthew Jennejohn, *The Transactional Dynamics of Market Fragility*, 85 L. & CONTEMP. PROBS. 281 (2022); Timothy Meyer & Ganesh Sitaraman, *The National Security Consequences of the Major Questions Doctrine*, 122 MICH. L. REV. 55, 76-77 (2023).

⁵⁵ Doni Bloomfield & Jeff Gordon, *The Law and Economics of Resilience*, 103 WASH. U.L. REV. 13-23 (forthcoming 2026).

⁵⁶ *Id.*, at 22-23.

risk itself for a firm's operation and directly contribute to the vulnerability of firms.⁵⁷ Some other scholarship addressed the effects of public policies on firm's capacities.⁵⁸ Public policies could draw favorable conditions or advantages to incentivize and stimulate, or they act as inhibiting such process. While this line of literature is vast, there are scarce ones addressing when the governmental interventions are optimal or should only rely on market dynamics to generate desired societal outcomes. Our investigation in the last section showed that some literature provided the variants to company's sufficient supply chain investment, others are more confirmed that resilience investment is lacking. They offered different policy considerations for governments to intervene.

Grossman, Helpman and Sabal maintained that as the first-best package, three types of subsidies shall be granted: subsidies (or taxes) on inter-firm transactions to neutralize bargaining markups; subsidies to firm-level resilience investments to internalize downstream externalities; and subsidies to supplier-network formation. When transaction subsidies are available, optimal resilience and network subsidies depend only on local technological and bargaining power at the targeted tier. However, subsidies to counter bargaining markups might not always be obtainable, the authors argue as the second-best approach, the amount of the subsidies shall be determined according to the location of the targeted industry: upstream firms typically warrant larger resilience subsidies than downstream firms, especially when downstream buyers' bargaining power is weakly declining along the chain. Upstream disruptions effects more widely, affecting multiple tiers and amplifying welfare losses through compounded markups.⁵⁹

Grossman, Helpman and Lhuillier examined whether governments should promote international diversification of supply chains or encourage reshoring. They found that whether governments shall intervene in diversification or reshoring depends on the demand structure used. They argue that under the constant elasticity of substitution (CES), consumers' willingness to switch away from a product when its price rises are

⁵⁷ See generally, Shardul Phadnis & Nitin Joglekar, *Configuring Supply Chain Dyads for Regulatory Disruptions: A Behavioral Study of Scenarios*, 30 PROD. & OPERATIONS MGMT. 1014 (2021); Jessica L. Darby et al., *The Implications of Firm-Specific Policy Risk, Policy Uncertainty, and Industry Factors for Inventory: A Resource Dependence Perspective*, 56 J. SUPPLY CHAIN MGMT. 3 (2020).

⁵⁸ E.g. Zuojun Max Shen & Yiqi Sun, *Strengthening supply chain resilience during COVID-19: A case study of JD.com*, 69 J. OPERATIONS MGMT. 359 (2023); Juan Camilo Serpa & Harish Krishnan, *Policy Incentives for Dangerous (But Necessary) Operations*, 25 PROD. & OPERATIONS MGMT. 1778 (2016); M. Douglas Voss & Zachary Williams, *Public-Private Partnerships and Supply Chain Security: C-TPAT as an Indicator of Relational Security*, 34 J. BUS. LOGISTICS 320 (2013).

⁵⁹ Grossman, Helpman & Sabal, *supra* note 43, at 2381-83, 2396.

always the same, no matter how expensive it becomes. When disruption occurs, the increase in consumer surplus from an additional available variety exactly offsets the business-stealing losses imposed on other firms, causing private profit gains to coincide with social welfare gains. However, constant elasticity of substitution is too idealized, Marshall's Second Law of Demand (MSLD) shall apply, under which consumers become more sensitive when prices rise. In this case, surviving firms earn disproportionately large profits in high-price disruption states, while the incremental consumer-surplus gains from additional availability are comparatively smaller since the consumers would cut their purchases sharply. This divergence causes private incentives for diversification and resilience to exceed their social value.⁶⁰

Grossman, Helpman and Lhuillier concluded that under CES preference, if consumption subsidies are available (hinges upon the government's power to fix price), no intervention shall be applied. If consumption subsidies are unavailable in the second-best environment, diversification shall be subsidized to create more source to lower markups, but shall be neutral between home and foreign sourcing. Under the MSLD cases, different consumption subsidies are needed for different products, accompanied with discouragement (through taxation) of diversification. If consumption subsidies are unavailable, the policy shall be qualitatively similar to the ones that achieve with consumption subsidies if the elasticity of demand is small, but lower taxes or even subsidies if the demand elasticity is large. With regards to the decision of reshoring, friend-shoring, or onshoring, policy shall be devised to offshore from risky and costly either through taxation if the elasticity of demand is small or subsidization if the demand elasticity is large.⁶¹

Bloomfield and Gordon discussed law's role in enhancing supply chain resilience in domestic context. They argued that dedicated resilience regulation is often the more optimal tool because it is explicitly about resilience, while business law has lots of other goals and can create messy tradeoffs. They identify three general categories of measures. First, quantity targeting mandates or procures *ex ante* capacity. These systems can overshoot because forecasting is difficult and imposes a broader knowledge and coordination burden on administrators. Second, price targeting is a less intrusive alternative that tries to sustain production by keeping prices high enough to support continued supply. This approach is simpler to administer but less precise, and it depends heavily on credibility. Third, stockpiles shift the burden to public inventories, with government storing and

⁶⁰ Grossman, Helpman & Lhuillier, *supra* note 46, at 3462, 3465-66.

⁶¹ *Id.*, at 3465.

releasing critical goods. The state may hold advantages through scale, lower capital costs, and the social insurance value of buffers, but repeated intervention can create collective moral hazard if market participants come to rely on public backstops in systemic shortages.⁶²

D. Takeaways from Economy Rationalities

The main takeaway from the prior inquiry is that no single, comprehensive policy package is universally optimal across products, sectors, and tiers of production networks. The economic literature goes beyond the claim that resilience is underprovided. It shows that private incentives can diverge from social welfare for multiple reasons, and that the direction of the distortion depends on structural features such as chain position, bargaining power, market concentration, input substitutability, and the shape of demand under disruption. The policy implication is not that subsidies are always desirable, but that the correct baseline is heterogeneity for subsidy design.

When resilience externalities dominate, the case for intervention is often strongest in upstream industries. Resilience investments by upstream firms can preserve the feasibility of production for multiple downstream users and reduce economy wide markups triggered by shortages. In these settings, supply side support for resilience or capacity can be welfare enhancing even when firms have rational incentives to underinvest. When business stealing and scarcity pricing dominate, undifferentiated diversification, reshoring, or redundancy initiatives can induce duplicative investments whose private returns exceed their social value. In such cases, blanket reshoring mandates, including measures that permanently ring fence particular source countries, and broad diversification subsidies are likely to be insensitive to demand structure and input substitutability.

These conclusions align with the observable pattern of contemporary resilience programs discussed in Part III. Jurisdictions rarely subsidize all supply chains equally. Instead, they concentrate support on a limited set of critical sectors, particularly those marked by strong upstream leverage and weak substitutability, including critical minerals, semiconductors, and certain medical products. They also tend to rely on case by case grant making, conditionality, and targeted procurement, rather than permanent, across the board exclusions of particular sourcing countries.

III. TYPOLOGY OF DISCRIMINATION IN SUBSIDIZATION

⁶² Bloomfield & Gordon, *supra* note 55, at 24-30.

Except for a limited set of aspirational or institution building instruments, trade law, particularly the disciplines on subsidies and countervailing measures, does not address supply chain resilience in direct terms. This Part therefore develops a typology of current state practice using trade law categories.

Resilience-oriented subsidies can be organized along two dimensions. The first concerns the subsidy's policy objective. A measure may seek to promote onshoring by attracting investment back to domestic production, or it may seek to promote reshoring or diversification by encouraging domestic firms to relocate or expand investment across foreign production sites or to specific sites deemed less risky than the current one. As a matter of trade law, these objectives map onto the GATT's MFN and national treatment obligations.⁶³ The second dimension concerns where, along the supply chain, selectivity is applied. A subsidy can confer advantages on particular inputs or on final products. Input based measures are often granted selectively to producers or commercial buyers that use domestic inputs rather than foreign inputs, or that use specified foreign inputs over other foreign inputs.⁶⁴ The immediate beneficiaries are typically upstream products that compete directly with foreign counterparts. Measures directed at final products operate through conferring benefits only to domestic producers, or to selected foreign producers through lower taxes, preferential loans or the purchase and sell of goods, rendering these producers in a more favorable position compared to others.

Combining these two dimensions yields four categories of strategic selectivity. In the onshoring setting, measures take the form of domestic production subsidies and subsidies conditioned on the use of domestic content. In the reshoring setting, measures take the form of designated foreign production subsidies and subsidies conditioned on the use of designated foreign content. It shall be emphasized that all four subsidies can be effectuate directly or indirectly: directness refers to a supply-side subsidy that would boost the production of intermediate or final products; indirectness refers to a consumer-side subsidy that grants benefits to the final consumers that indirectly benefit the producer through generating more profits for them. This section illustrates each category through a stylized hypothetical involving three trading states, *A*, *B*, and *C*. Each state produces a competitive, substitutable product, *a*, *b*, and *c*. A common input, *X*, is required for all three products and is available in each state.

⁶³ GATT, arts. I:1, III.

⁶⁴ We use the terminology of producers/buyers in the same way as Alan O. Sykes discussing the rationale of GATT Article III:8(b). ALAN O. SYKES, *THE LAW AND ECONOMICS OF INTERNATIONAL TRADE AGREEMENTS* 396 (2023).

A. Onshoring Subsidies*1. Domestic Production Subsidies*

Domestic production subsidies describe subsidization provided exclusively to domestic producers, excluding foreign producers from accessing such grants. In our hypothetical case, State *B* and State *C* both export *b* and *c* into the market of State *A*, competing with product *a*. Such subsidies are granted exclusively to producers from state *A* in the production of *a*, with no subsidization extended to producers from States *B* or *C*. Such subsidies may also be implemented indirectly, through tax credits granted to consumers purchasing domestic products. The exclusive provision of domestic producer subsidies can be motivated the protection of infant industry,⁶⁵ or raising domestic welfare by shifting profits from foreign to domestic companies.⁶⁶ This rationale dates back to the ITO Charter, under which negotiators “bifurcated” export and domestic subsidies, restricting only the former, while leaving broad policy discretion for the latter.⁶⁷

Recent practice has been triumphed by “Made in X” programs by various states before Covid-19,⁶⁸ including the “Blueprint for an America Built to Last” in 2012 during the Obama Administration,⁶⁹ Korea’s reshoring subsidies and tax reduction in 2013,⁷⁰ “Reshoring UK” initiative in 2014,⁷¹ etc. Kenya and Nigeria respectively initiated its programs to promote local production and reduce import dependence. Particularly, the “Buy Kenya, Build Kenya” promotes that 40% of the value of government procurement be sourced locally. Tax incentives (such as rebates on advertising locally made goods) and preferential access to public tenders are used to encourage Kenyan manufacturing.⁷² China

⁶⁵ Marc J. Melitz, *When and How Should Infant Industries Be Protected?*, 66 J. INT’L ECON. 117, 118 (2005).

⁶⁶ W. MAX. CORDEN, *TRADE POLICY AND ECONOMIC WELFARE* 244 (2nd ed. 2003).

⁶⁷ PETROS C. MAVROIDIS, *THE REGULATION OF INTERNATIONAL TRADE, VOLUME 2: THE WTO AGREEMENTS ON TRADE IN GOODS* 186 (2016).

⁶⁸ See generally, Stefano Elia et al., *Post-Pandemic Reconfiguration from Global to Domestic and Regional Value Chains: The Role of Industrial Policies*, 28 TRANSNAT’L CORP. 67, 75-78 (2021) (discussing chronically various state’s industrial policies to propel reshoring).

⁶⁹ *Blueprint for an America Built to Last*, The White House (Jan. 24, 2012), https://obamawhitehouse.archives.gov/sites/default/files/blueprint_for_an_america_built_to_last.pdf.

⁷⁰ Haeoejinchulgi-eop-ui guknaebokgwi jiwone gwanhan beobyul [Act on Assistance to Korean Off-Shore Enterprises in Repatriation to Korea] arts. 11-12 (S. Kor.).

⁷¹ Dep’t Bus., Innovation & Skills, *New Government Support to Encourage Manufacturing Production Back to the UK* (Jan. 24, 2014), <https://www.gov.uk/government/news/new-government-support-to-encourage-manufacturing-production-back-to-the-uk>.

⁷² Kenya Inst. Pub. Pol’y Rsch. & Analysis, *Unlocking the Potential for Domestic Market to Thrive: A Case of Buy Kenya Build Kenya* (Jan. 8, 2024), <https://kippra.or.ke/unlocking->

has probably provided one of the largest amounts of support through its “Made in China 2025” initiative.⁷³ The ambitious regime sought to transform China into a high-tech manufacturing powerhouse, aiming to achieve 70 percent self-sufficiency in high-tech technology. The Chinese government provided subsidies, state investments, and low-interest loans to firms in the targeted industries. Tax breaks and financial incentives were offered to scale up production in areas like chips, electric vehicles and medical devices.⁷⁴

Post-pandemic industrial policy continues the previous trend of propelling domestic production. Some were propelled by the reshoring of pharmaceuticals and medical kits production.⁷⁵ Later ones help to promote sustainable development, including the EU’s Net-Zero Industry Act with the target of at least 40 percent of its clean-technology needs (such as solar panels, wind turbines, batteries, heat pumps) domestically by 2030.⁷⁶

The U.S. practice sustain a unique practice of inquiry. The U.S. grant certain commonly observed domestic production subsidies. For instance, after repealing the IRA’s clean vehicle credit, the Trump administration substitutes it with another tax deduction regime under the One Big Beautiful Bill benefiting only domestic automobile producers. US tax law prohibits a deduction for personal interest paid or accrued during the taxable year in calculating taxable income.⁷⁷ The Bill provides a special rule for taxable years 2025 through 2028 that allows personal interest paid on qualified passenger vehicle loans could be deductible. Such qualified passenger vehicles shall “not include any vehicle the final assembly of which did not occur within the U.S.”⁷⁸ Though the subsidy is not directly conferred on the producers or assemblers of the automobiles, producers in the US could potentially receive competitive benefits, as the consumers will be prone to purchase cars with better loaning policies and save on paying taxes.

Eligibility criteria for domestic production subsidies increasingly include

the-potential-for-domestic-market-to-thrive-a-case-of-buy-kenya-build-kenya/; Executive Order (On Support for Local Content in Public Procurement by Ministries, Department and Agencies of the Federal Government of Nigeria) (2017) 104:55 O.G., A25 (Nigeria).

⁷³ Guowuyuan Guanyu Yinfā “Zhongguo Zhizao 2025” de Tongzhi (国务院关于印发《中国制造 2025》的通知) [Notice of the State Council on Issuing “Made in China 2025”] (promulgated by the State Council, May 19, 2015, effective May 19, 2015).

⁷⁴ Council on Foreign Relations, *Is “Made in China 2025” a Threat to Global Trade?*, (Aug. 2, 2018), <https://www.cfr.org/backgrounders/made-china-2025-threat-global-trade>.

⁷⁵ See discussion by Mitchell and Brow, *supra* notes 33, 34, 35.

⁷⁶ Regulation (EU) 2024/1735 of the European Parliament and of the Council of 13 June 2024 on Establishing a Framework of Measures for Strengthening Europe’s Net-Zero Technology Manufacturing Ecosystem (Net-Zero Industry Act) and Amending Regulation (EU) 2018/1724 art. 5(1)(a), 2024 O.J. (L 1735) 1 (June 28, 2024)

⁷⁷ 26 U.S.C. § 163(h) (2025).

⁷⁸ One Big Beautiful Bill, § 70203 (2025).

requirements tied to a firm's ownership and control structure. The CHIPS and Science Act during the Biden administration showcases such a practice.⁷⁹ The regime bars any "foreign entities of concern" from being eligible for the fundings of establishing and expanding production facilities within the U.S. These entities include those owned by, controlled by, or subject to the direction and jurisdiction of a covered nation.⁸⁰ In this setting, strategic selectivity extends beyond the onshoring location decision and also turns on which foreign participants may be welcomed into domestic production.

2. *Subsidies Contingent upon Using Domestic Content*

Subsidies contingent upon using domestic content discriminate against foreign content. In the hypothetical case, state *B* exports content *x* to state *A*, which also produces content *x* domestically. Such subsidies are granted when State *A* adopts a regime under which only companies using domestically produced *x* are eligible for certain direct transfers of funds, tax credits, and other financial contributions, while companies procuring *x* from foreign producers are excluded from the benefits.

Subsidies contingent upon using domestic content have been challenged under the WTO dispute settlement system in seven disputes. Relevant measures include: manufacture grants conditioned upon "counter-purchasing" local car parts and components; import tax exemption tied with local value added in vehicles production; payment program contingent on the use of domestic upland cotton; feed-in tariff program that required a "minimum required domestic content level;" support for Airbus contingent on the production of "domestic" aircraft goods; aerospace tax granted conditioning on using local-made goods; and tax exemptions for information and communication technology programs and automotive supply chain innovation, conditioned upon using domestic products.⁸¹

Subsidies contingent upon using domestic content might result in inefficient allocation of resources.⁸² Proponents of the scheme argue that prohibition of domestic content subsidies neglects the possible spill-over effect (especially in mitigating climate change) from the calculation of overall welfare.⁸³ However, the prohibition of domestic content

⁷⁹ William M. (Mac) Thornberry National Defense Authorization Act for Fiscal Year 2021, Pub. L. No. 116-283, § 9907, 134 Stat. 3388, 4860 (2022).

⁸⁰ Infrastructure Investment and Jobs Act, Pub. L. No. 117-58, sec. 40207(a)(5), 135 Stat. 429, 963 (2021) (as codified in 42 U.S.C. sec. 18741(a)(5) (2025)). "Cover nations" include North Korea, China, Russia and Iran. 10 U.S.C. sec. 4872(d)(2) (2025).

⁸¹ These seven cases involve: *Indonesia – Autos*, *Canada – Autos*; *US – Upland Cotton*, *Canada – Feed-In Tariff*, *EC – Aircraft*, *US – Tax Incentives*, and *Brazil – Taxation*.

⁸² Mark Wu & James Salzman, *The Next Generation of Trade and Environment Conflicts: The Rise of Green Industrial Policy*, 108 NW. U. L. REV. 401, 423 (2014).

⁸³ Timothy Meyer, *How Local Discrimination Can Promote Global Public Goods*, 95 B.U.

subsidies is the reflection of the NT in the context of the subsidy regulations and seeks to prevent trade distortion resulting from the explicit design to replace imports and to negate the effect of tariff concessions.⁸⁴

B. Friend-Shoring Subsidies

The onshoring subsidies seems familiar to any trade lawyers. Under the WTO regime, policy makers are well cautious about the potential violations and be labelled as protectionists. However, friend-shoring subsidies seem more newly formed and generally relates to a state's impulse to channel productions to less geopolitically risky states or to expand and diversify their production in other areas.

1. Designated Foreign Production Subsidies

Designated foreign production subsidies are provided exclusively to certain foreign producers to conduct production in the beneficiary's home state or the subsidizing state, while excluding the competing producers from other foreign states. The latter is rarely granted in isolation; they typically accompany domestic production subsidies. In our hypothetical case, States *B* and *C* both export *b* and *c* into the market of State *A*, competing with product *a*. Subsidies are provided for the production of *a* and *b*, while producers from State *C* receive no subsidies. Such support might also be provided indirectly through tax credits granted to consumers purchasing *a* and *b*.

EU has widely adopted subsidies and provide investment funds for designated foreign production. The EU Critical Raw Materials Act sets the ceiling that no more than 65% dependency on a single third country for any strategic raw material.⁸⁵ To reach this objective, the EU initiate public investment for alternative sourcing, with the priority for states establishing Strategic Partnership by implementing fast-track permission and financing for projects within these states to reduce single-country dependency.⁸⁶ By far, the EU has funded 47 in-EU projects and 13 out-of-EU projects in Ukraine, New Caledonia, Greenland, Norway, etc. South Korea also initiated its overseas supply-chain investment, including equity investment in foreign firms that produce economic-security items or related services and critical mineral investment.⁸⁷

L. REV. 1939, 1986 (2015).

⁸⁴ MAVROIDIS, *supra* note 67, at 273.

⁸⁵ Regulation (EU) 2024/1252 of the European Parliament and of the Council of 13 June 2024 on Establishing a Framework of Measures for Strengthening Europe's Critical Raw Materials Value Chain and Amending Regulation (EU) 2018/1724 art. 5(1)(b), 2024 O.J.

⁸⁶ *Id.*

⁸⁷ *Selected Strategic Projects*, EU. COMM'N, <https://single-market->

In 2020, Japan also initiated its Overseas Supply Chain Diversification Support Project. The project aimed to relief its reliance on production and imports from China and stimulated the introduction of facilities and equipment in ASEAN countries.⁸⁸ However, this project only provided financial support for private companies which have a business base, activity and legal entity status in Japan.⁸⁹ The last invitation for the program ended in 2023. Immediately afterwards, the 2023 amendment of the Japan Bank of International Cooperation Act included foreign companies to be eligible for loans if they are included in the supply chains and industry infrastructure of Japan. The application process shall be made by a Japanese company, including overseas Japanese affiliates.⁹⁰ The Bank will assess whether financing is necessary to strengthen supply chains or critical technologies for Japanese companies, to avoid overdependence on specific countries, to support transactions where Japanese firms are primary suppliers or face limited alternatives, to facilitate production expansion benefiting Japanese buyers.⁹¹ In practice, the loan is generally provided to Japanese companies' foreign subsidiaries, including those in the U.S., China, Australia, Brazil.⁹² Several loans are provided to fully owned foreign companies in

economy.ec.europa.eu/sectors/raw-materials/areas-specific-interest/critical-raw-materials/strategic-projects-under-crma/selected-projects_en, (last visited Feb. 24, 2026).

⁸⁸ *Overseas Supply Chain Diversification and Other Support Projects*, JAPAN EXTERNAL TRADE ORG., <https://www.jetro.go.jp/services/supplychain/?utm> (last visited Feb. 24, 2026).

⁸⁹ *Support Program to Strengthen Overseas Supply Chains For Demonstration Projects and Project Feasibility Studies*, JAPAN EXTERNAL TRADE ORG., https://www.jetro.go.jp/ext_images/services/supplychain/info-2/2nd_en.pdf (last visited Feb. 24, 2026).

⁹⁰ Kabushiki Kaisha Kokusai Kyōryoku Ginkō Hō [Japan Bank for International Cooperation Act], Law No. 39 of 2011 (Japan), as amended by Kabushiki Kaisha Kokusai Kyōryoku Ginkō Hō no Ichibu o Kaisei Suru Hōritsu [Act for Partial Amendment of the Japan Bank for International Cooperation Act], Law No. 11 of 2023, arts. 2(10)(a)-(b), 11(6)(4) (Japan); Kabushiki Kaisha Kokusai Kyōryoku Ginkō Hō Shikō Kisoku [Regulations for Enforcement of the Japan Bank for International Cooperation Act], Ministry of Finance Ordinance No. 14 of 2012, art. 4(1) (Japan).

⁹¹ Enhancement of Supply Chain Resilience to Contribute to Strengthening Japan's Industrial Competitiveness: 023 JBIC Act Amendment: Three Key Points, JAPAN BANK INT'L COOP., https://www.jbic.go.jp/en/information/today/today_2023sp/jtd_202312_sp4.html (last visited Feb. 24, 2026).

⁹² *Loan for BEV Battery Manufacturing Business of Toyota Motor Corporation's U.S. Subsidiary*, JAPAN BANK INT'L COOP. (May 28, 2024), https://www.jbic.go.jp/en/information/press/press-2024/press_00021.html; *Financing for the Manufacturing and Sales of Printed Circuit Boards Carried out by Japanese Companies in China*, JAPAN BANK INT'L COOP. (Mar. 14, 2025), https://www.jbic.go.jp/ja/information/press/press-2024/press_00148.html; *Loan for Acquisition of Coregas Group by Nippon Sanso Holdings Corporation*, JAPAN BANK INT'L COOP. (Dec. 19, 2025), <https://www.jbic.go.jp/en/information/press/press->

India, Singapore, Germany, and Chile to secure Japanese access to critical goods.⁹³

2. *Subsidies Contingent upon Using Designated Foreign Content*

Subsidies contingent upon using designated foreign content are provided with the requirement that inputs originating from certain designated foreign states be used in preference to those from other foreign states. In our hypothetical case, states *B* and *C* both export *x* to state *A*. Such discriminatory subsidies are granted when state *A* devises a subsidy regime under which only companies that use *x* produced in State *B*, or that avoid using *x* from State *C*, are eligible to receive direct transfers of funds, tax credits, or other forms of financial contribution.

The expired clean vehicle credit under the IRA is a typical example of a hybrid model of both domestic content and foreign content subsidy. There is a two-tier requirement for the clean vehicle credit eligibility.⁹⁴ First, the qualified clean vehicle must be one “the final assembly of which occurs within North America.” Second, a credit of \$3,750 is granted if either the critical minerals or battery components requirement is satisfied, and \$7,500 if both are met. The critical minerals requirement stipulates a minimum battery critical minerals value to be extracted or processed in the U.S., in a country maintaining an FTA with the US, or recycled in North America. The battery components requirement similarly mandates a minimum battery components value to be manufactured or assembled in North America. Furthermore, if the battery critical minerals or the battery components are sourced from or recycled in a foreign entity of concern, the vehicle would be automatically disqualified. These provisions effectively exclude inputs from China, Russia, Iran, and North Korea.⁹⁵

Before expiration, eligible vehicles include not only U.S. brands such as Tesla and Cadillac, but also foreign-made cars by manufacturers like Honda, KIA, and Acura.⁹⁶

2025/press_00134.html; *Loan to ALBRAS - Alumínio Brasileiro S.A. in Brazil*, JAPAN BANK INT’L COOP. (Oct. 29, 2025), https://www.jbic.go.jp/en/information/press/press-2025/press_00103.html

⁹³ *Untied Loan to IndusInd Bank Limited of India*, JAPAN BANK INT’L COOP. (Mar. 6, 2024), https://www.jbic.go.jp/en/information/press/press-2023/press_00194.html; *Loan to Trafigura Pte Ltd.*, JAPAN BANK INT’L COOP. (Mar. 29, 2024), https://www.jbic.go.jp/en/information/press/press-2023/press_00218.html; *Loan to United Internet AG of Germany*, JAPAN BANK INT’L COOP.. (Dec. 27, 2024), https://www.jbic.go.jp/en/information/press/press-2024/press_00116.html; *Loan to Corporación Nacional del Cobre de Chile*, JAPAN BANK INT’L COOP. (Apr. 1, 2025), https://www.jbic.go.jp/en/information/press/press-2025/press_00003.html.

⁹⁴ IRA, § 13401 (as codified in 26 U.S.C. § 30D (2025)).

⁹⁵ Infrastructure Investment and Jobs Act, § 40207(a)(5).

⁹⁶ *Federal Tax Credits for New Plug-in Electric and Fuel Cell Electric Vehicles Purchased*

Discrimination exists between the upstream raw materials originating from the North American states and other states signing FTAs with the U.S., on the one hand, and those originating from China, on the other. Such discrimination is driven by two underlying initiatives. Selective subsidization often operates in tandem with trade measures, including tariffs, to “protect, or ‘ring fence,’ domestic industrial policy strategies from foreign competition,” thereby shielding emerging domestic sectors.⁹⁷ Moreover, it aligned with the U.S.’s then strategy to enhance supply chain resilience in critical minerals. This strategy has led to the formation of frameworks and partnerships such as the Indo-Pacific Economic Framework for Prosperity, the Mineral Security Partnership, and various bilateral or plurilateral critical mineral agreements negotiated with allies and trade partners.

Another example can be found in South Korea’s supply chain stabilization support regime. In its 2024 First Supply Chain Stabilization Master Plan for the year 2025 to 2027, fundings are provided for third-country import: price-difference compensation or transport-cost subsidies when firms import from a third country instead of the dominant supplier for items with high dependence on a particular country.⁹⁸ These two import diversification measures are granted from the Supply Chain Stabilization Fund.⁹⁹ Additional support includes import insurance for these diversified imports and end-to-end consulting support for small and medium-sized enterprises to locate other import sources.¹⁰⁰ The 2025 annual Supply Chain Stabilization Implementation Plan further details the price-difference compensation or transport-cost subsidies.¹⁰¹ The Plan reported that in 2024, 50% of the freight cost for vehicle-use urea, a nitrogen compound used in diesel vehicles to reduce NOx emissions, were subsidized for third-state procurement instead from China by the Ministry of Industry. ₩3 billion were expected to be provided in 2025 to support long-term

in 2023 or After, US DEP’T OF ENERGY, <https://fueleconomy.gov/feg/tax2023.shtml> (last visited Mar. 12, 2025).

⁹⁷ William Tobin, *The IRA Two Years On: A Signpost of the New Economic Policy Consensus*, ATL. COUNCIL (Aug. 15, 2024), <https://www.atlanticcouncil.org/blogs/new-atlanticist/the-ira-two-years-on-a-signpost-of-the-new-economic-policy-consensus/>.

⁹⁸ Je Icha Gonggeummang Anjeonghwa Gibon Gyehoek [*First Supply Chain Stabilization Master Plan (2025-2027)*], S. KOR. MIN. FIN. ECON. 20-23 (Dec. 19, 2024), https://mofe.go.kr/com/synap/synapView.do;jsessionid=BTfNEC6yk87MlywHULxUY6UPZ4TXwDzfrnftJVqB.node40?atchFileId=ATCH_000000000027925&fileSn=4.

⁹⁹ Framework Act on Supply Chain Stabilization Support for Economic Security, *supra* note 20, arts. 22, 38.

¹⁰⁰ *First Supply Chain Stabilization Master Plan (2025-2027)*, *supra* note 98, at 23.

¹⁰¹ 2025 Gonggeummang Anjeonghwa Sihaeng Gyehoek [*2025 Supply Chain Stabilization Implementation Plan*], S. KOR. MIN. FIN. ECON. 19, 23 (Mar. 25, 2025), https://mofe.go.kr/com/synap/synapView.do?atchFileId=ATCH_000000000028599&fileSn=13.

import contract costs with third countries.¹⁰² Additionally, other supply stability support programs are expected to be established and budgeted for high-risk economic security items and sources, including semiconductors, AI, batteries, critical minerals, aviation, defense, and essential consumer goods.¹⁰³

IV. LEGITIMACY UNDER GROUND RULES OF WTO

Part III surveys contemporary state practice in using subsidies to strengthen supply chain resilience. This part evaluates the legality of these measures under the WTO ground rules. Despite the growing prominence of resilience as a policy objective and the proliferation of domestic and cooperative initiatives, WTO law remains the prevailing baseline for disciplining subsidization in international trade. Its rules have sweeping effect because they structure both multilateral challenges and unilateral responses, including countervailing investigations and dispute settlement claims. In practice, Members design, justify, and contest subsidy programs with the WTO framework in view, even when no dispute is immediately foreseeable. This Part therefore examines resilience-oriented subsidies against the disciplines of the SCM Agreement and the GATT, and it assesses how the relevant obligations and exceptions shape the legal space for selective support measures.

A. Legitimacy Under the SCM Agreement

Article 1.1 of the SCM Agreement first prescribes that a subsidy is constituted of a financial contribution by a government or any public body in the form of: (1) a direct transfer of funds (e.g. grants, loans, and equity infusion), potential direct transfers of funds or liabilities (e.g. loan guarantees); (2) government revenue that is otherwise due is foregone or not collected (e.g. fiscal incentives such as tax credits); (3) a governmental provision of goods or services other than general infrastructure, or governmental purchases of goods; (4) governmental payments to a funding mechanism, or entrusting or direction of a private body to carry out above functions. Moreover, any form of income or price support in the sense of Article XVI of GATT also constitutes a subsidy.¹⁰⁴

The SCM Agreement then trifurcates subsidies into prohibited, actionable, and non-actionable subsidies, each subject to distinct legal disciplines. This tripartite structure reflects the SCM Agreement's central aim of balancing Members' policy space to provide

¹⁰² *Id.*

¹⁰³ *Id.*, at 34, 76-79.

¹⁰⁴ SCM Agreement art. 1.1.

financial support with the imperative of curbing trade-distorting effects. First, prohibited subsidies are deemed *per se* inconsistent with WTO rules due to their inherently trade-distortive nature and include subsidies contingent, in law or in fact, either upon export performance or the use of domestic over imported goods.¹⁰⁵ Such subsidies are presumed to distort trade and must be withdrawn without the need for demonstrating adverse effects. Second, actionable subsidies, the default category, are not prohibited *per se* but may be challenged if they cause adverse effects to the interests of other Members.¹⁰⁶ These adverse effects include injury to the domestic industry of another Member, nullification or impairment of benefits accruing under the GATT, or serious prejudice to the interests of another Member.¹⁰⁷ Demonstrating these effects requires rigorous factual and legal analysis, and the burden rests on the complaining Member. Third, the category of non-actionable subsidies originally encompassed certain narrowly defined programs presumed not to cause adverse trade effects, such as subsidies for research activities, disadvantaged regions, or environmental compliance.¹⁰⁸ However, the provisions authorizing this category lapsed in 2000 and were not renewed, leaving only the prohibited and actionable categories in force today.

Article 3.1(b) of the SCM Agreement can discipline reshoring subsidies when eligibility is contingent on the use of domestic content. The prohibition on domestic content subsidies reflects national treatment concerns within subsidy regulation and aims to prevent trade distortion that results from the explicit substitution of imported raw materials or inputs with domestic ones.¹⁰⁹ As the U.S. delegate argued during the Uruguay Round, subsidies contingent upon using domestic content “are as effective as any tariff in protecting domestic input supplying industries and distorting the flow of resources internationally.”¹¹⁰ However, for other types of subsidies, they are not subject to a blatant prohibition because the SCM Agreement does not expressly regulate strategic selectivity as such. Instead, selectivity operates through the specificity requirement, which determines whether a subsidy is actionable.

¹⁰⁵ SCM Agreement art. 3.

¹⁰⁶ SCM Agreement art. 5.

¹⁰⁷ *Id.*

¹⁰⁸ SCM Agreement art. 8.

¹⁰⁹ MAVROIDIS, *supra* note 67, 273.

¹¹⁰ Group of Negotiations on Goods (GATT) Negotiation Group on Subsidies and Countervailing Duties, *Elements of the Framework for Negotiations: Submission by the US*, WTO Doc. MTN.GNG/NG10/W/29, 2 (Nov. 22, 1989); M. Jean Anderson & Gregory Husisian, *The Subsidies Agreement*, in *THE WORLD TRADE ORGANIZATION: THE MULTILATERAL TRADE FRAMEWORK FOR THE 21ST CENTURY AND U.S. IMPLEMENTING LEGISLATION* 299, 309 (Terence P. Stewart ed., 1996).

Accordingly, other types of resilience oriented subsidies may still be challenged and countervailed if they cause adverse effects under Article 5. Article 5 identifies three forms of adverse effects: “injury to the domestic industry of another Member,” “nullification or impairment of benefits accruing directly or indirectly to other Members under GATT 1994 in particular the benefits of concessions bound under Article II of GATT 1994,” and “serious prejudice to the interests of another Member.”¹¹¹ “Serious prejudice” includes displacement or impediment of another Member’s imports of like products in the subsidizing Member’s market or in a third country market, significant price undercutting or suppression of a like product in the same market, or an increase in the subsidizing Member’s world market share in a subsidized primary product.¹¹²

Domestic production subsidies designed to support resilience and supply continuity during crises are less likely to generate injury to another Member’s domestic industry when the subsidized products are not exported in significant quantities. They may nonetheless cause serious prejudice through displacement or impediment of another Member’s like product imports in the subsidizing Member’s market. Subsidies contingent on the use of designated foreign content can raise similar concerns, because they can steer demand away from inputs originating in excluded jurisdictions and thereby impede the competitive opportunities of those inputs and their downstream products in the subsidizing market.

Foreign production subsidies raise a different threshold issue. Friend-shoring subsidies support production of goods or raw materials in specified foreign locations for importation into the subsidizing Member for consumption or downstream manufacturing. This practice raises the question whether the SCM Agreement contains a territorial limitation on the subsidies it regulates. Article 1.1(a)(1) refers to “a financial contribution by a government or any public body within the territory of a Member.”¹¹³ This language is often read as limiting the identity of the provider rather than the location of the beneficiary, as reflected in common export subsidy practices.¹¹⁴ The more significant hurdle arises under Article 2, which ties specificity to subsidies provided to enterprises or industries “within the jurisdiction of the granting authority.”¹¹⁵ The European

¹¹¹ SCM Agreement art. 5.

¹¹² SCM Agreement arts. 6.1(c), 6.3(a)-(d).

¹¹³ SCM Agreement art. 1.1(a)(1).

¹¹⁴ Victor Crochet & Vineet Hegde, *China’s “Going Global” Policy: Transnational Production Subsidies Under the WTO SCM Agreement*, 23 J. INT’L ECON. L. 841, 847 (2020); Victor Crochet & Marcus Gustafsson, *Lawful Remedy or Illegal Response? Resolving the Issue of Foreign Subsidization under WTO Law*, 20 WORLD TRADE REV. 343, 348-49 (2021); Weihuan Zhou & Victor Crochet, *Attribution to the State: A Critique of Cross-Border Subsidies*, 25 MELB. J. INT’L L. 2, 15-16 (2024).

¹¹⁵ Crochet & Hegde, *supra* note 114, at 849-51; Crochet & Gustafsson, *supra* note 114, at

Commission's *Glass Fiber Fabrics* countervailing determination sought to address this issue by attributing subsidies provided by "Chinese public bodies" to the Egyptian government and then analyzing the jurisdictional link to Egypt.¹¹⁶ The argument then turns on principles of attribution and state responsibility under general international law, a move that WTO panels have treated with caution, including in *EU—CVD/AD on SSCRFP (Indonesia)*.¹¹⁷

B. Legitimacy Under the GATT

Subsidies aimed at strengthening supply chain resilience often operate through selectivity among origins or host locations, and they can therefore raise concerns under the MFN and national treatment obligations in Articles I:1 and III of the GATT.¹¹⁸ These

350; Ru Ding & Yiqian Xiang, *Emerging Rules on Cross-Border Subsidies: A Typological Analysis and Proposals for China's Approach*, 12 J. WTO & China 45, 50-52 (2022).

¹¹⁶ Commission Implementing Regulation (EU) 2020/776 of 12 June 2020 Imposing Definitive Countervailing Duties on Imports of Certain Woven and/or Stitched Glass Fibre Fabrics Originating in the People's Republic of China and Egypt and Amending Commission Implementing Regulation (EU) 2020/492 Imposing Definitive Anti-Dumping Duties on Imports of Certain Woven and/or Stitched Glass Fibre Fabrics Originating in the People's Republic of China and Egypt, 2020 O.J. (L 189). For comments on this issue, see generally, Zaker Ahmad, *The European Commission's Glass Fibre Fabrics Investigation and the Boundaries Between Investment and Trade*, 14 ASIA J. INT'L L. 72 (2024); Zhou & Crochet, *supra* note 114, at 9-15.

¹¹⁷ Panel Report, *European Union—Countervailing and Anti-Dumping Duties on Stainless Steel Cold-Rolled Flat Products from Indonesia*, ¶¶ 7.101-104, WTO Doc. WT/DS616/R (circulated Oct. 2, 2025). By far the question has been rendered moot since both the EU and Indonesia appealed this decision to the paralyzed Appellate Body. Due to the inapplicability of the WTO regime in sanctioning these outward investment subsidies (in our terms the friend-shoring subsidies), the EU issued its Foreign Subsidies Regulations seeking to tackle the so-called multilateral deficiency. See Marios Tokas, *Playing the Game: The EU's Proposed Regulation on Foreign Subsidies*, 56 J. WORLD TRADE 779 (2022); Malte Frank, *The EU's New Foreign Subsidy Regulation on Collision Course with the WTO*, 60 COMMON MKT. L. REV. 925 (2023). In addition to the potential violation of WTO rules as such, the authors hold the view that using this unilateral regime to counter our states' regime would be on itself a violation of the principle of estoppel, especially when the EU is expanding its outward foreign production subsidies in critical mineral areas.

¹¹⁸ There is limited scholarship focusing on this topic yet or only targeting limited scope of domestic content subsidies, e.g., Luca Rubini, *The Subsidization of Renewable Energy in the WTO: Issues and Perspectives* 28 (Nat'l Ctrs. Competence Res. Trade Reg., Working Paper No 2011/32), https://www.wti.org/media/filer_public/95/fe/95fe7812-471d-46ca-b1b2-523b3dbae005/2011-32_lr.pdf; Henok Asmelash, *The First Ten Years of WTO Jurisprudence on Renewable Energy Support Measures: Has the Dust Settled Yet?*, 21 WORLD TRADE REV. 455, 458 (2022). Doctrinal research is not catching up with the practical developments especially argued by China. China reprimanded the US's discriminatory subsidies before the Committee on Subsidies and Countervailing Measures. China's 2023 Report on WTO Compliance of the US also discussed the discrimination in subsidies granted contingent not only upon the use of domestic contents but also upon the use of certain foreign contents. Subsidies granted based on the nationality of the producer were also discussed. Committee on

measures may nonetheless be defended on several legal grounds, including the subsidy carve out in Article III:8(b),¹¹⁹ the general and security exceptions under Articles XX and XXI, and the free-trade areas exception under Article XXIV. This Part begins by identifying which resilience related subsidy measures fall within the scope of Articles I and III and then examines the availability and limits of these exceptions.

1. *Applicability of the Non-Discrimination Principle*

Articles I:1 and III have a broad coverage. Article III:2 applies to “internal taxes or other internal charges of any kind” on products and Article III:4 applies to “all laws, regulations and requirements affecting their internal sale, offering for sale, purchase, transportation, distribution or use.”¹²⁰ Article I:1 applies not only to “all matters referred to in paragraphs 2 and 4 of Article III,” but also to “customs duties and charges of any kind” and “the method of levying such duties and charges,” and “all rules and formalities in connection with importation and exportation.”¹²¹ While the scope of Article III:2 is more limited only to taxations measures, the WTO dispute settlement practice expounds Article III:4 a broader range of measures hinges on the terms “affecting,” even only focusing on certain element of the preferential taxation qualification.¹²² In *EC–Bananas III*, the Appellate Body ruled that “affecting” “implies a measure that has ‘an effect on,’ which indicates a broad scope of application.”¹²³

Subsidies and Countervailing Measures, *Minutes of the Regular Meeting Held on 27 October 2023*, 45-49, ¶¶ 365-391, WTO Doc. G/SCM/M/125 (Dec. 7, 2023); Ministry of Commerce of the People’s Republic of China 2023 Report on WTO Compliance of the United States, Aug. 11, 2023, 19-22. To the authors’ knowledge, only limited scholarship discussed the broader applicability of non-discrimination principle: DOMINIC COPPENS, WTO DISCIPLINES ON SUBSIDIES AND COUNTERVAILING MEASURES: BALANCING POLICY SPACE AND LEGAL CONSTRAINTS 189-192 (2014) (arguing national treatment does not apply to producer subsidies, we disagree on this broad construction, *see infra* part IV.B.1.b); Luca Rubini, *Subsidies, in HANDBOOK OF DEEP TRADE AGREEMENTS* 440 (Aaditya Mattoo, Nadia Rocha, & Michele Ruta eds., 2020) (discussed the application of non-discrimination obligations to subsidies under FTAs, we disagree also in part on the scope and sweeping effect of Article III:8(b) exception, *see infra* part V.C.3); *c.f.* Joel Slemrod & Reuven Avi-Yonah, *(How) Should Trade Agreements Deal With Income Tax Issues?*, 55 TAX L. REV. 533, 535-37 (2002) (arguing that a destination-based consumption tax could duplicate the effects of tariff and direct tax on producers as subsidies might have protectionism impact and should be subject to the non-discrimination obligation).

¹¹⁹ The exception occurs under the national treatment obligation, yet the Panel in *EC – Vessels* ruled that it also applies to the MFN obligation. Panel Report, *European Communities–Measures Affecting Trade in Commercial Vessels*, ¶¶ 7.84-90, WTO Doc. WT/DS301/R (adopted June 20, 2025) [hereinafter *EC–Vessels Panel Report*].

¹²⁰ GATT arts. III:2, III:4.

¹²¹ GATT art. I:1.

¹²² *See infra* discussion on the *U.S.–FSC* case.

¹²³ Appellate Body Report, *European Communities—Regime for the Importation, Sale and*

a. Content Contingency Subsidies Are Discriminatory

Referring back to our typology of supply chain resilience subsidies, subsidies contingent on the use of domestic or foreign content are particularly susceptible to challenges under national treatment and MFN. In the product market, the standard approach under Articles I:1 and III:4 examines whether the challenged measure alters the competitive relationship between domestic and foreign products, or among foreign products.¹²⁴ The Appellate Body in *Thailand–Cigarettes (Philippines)* indicated that the evaluation of the competitive relationship needed not to be “based on empirical evidence as to the actual effects of the measure at issue in the internal market,” but rather focus on the “the design, structure, and expected operation of the measure at issue” and the “genuine relationship between the measure at issue and its adverse impact on competitive opportunities.”¹²⁵

When implemented directly, content contingent subsidies affect competitive conditions between domestic and foreign inputs, or among foreign inputs, because downstream producers have incentives to purchase the designated inputs in order to obtain the subsidy. Similar effects can arise through indirect consumption side subsidies. Where final consumers receive support tied to the use of designated inputs, downstream producers are encouraged to incorporate those inputs in order to expand sales, since the final products become relatively cheaper to consumers. Accordingly, across forms, including income tax exemptions, reductions, or suspensions for producers, and reductions in indirect taxes borne by purchasers, such measures can undermine competitive opportunities for all foreign inputs, or for a subset of foreign inputs, in the relevant market.

b. Producer Subsidies Are Discrimination Only If There is Sufficient Producer-Product Linkage

Producer subsidies present a more intricate case. Where a subsidy operates indirectly through consumer side support, it typically falls within the ambit of Articles I and III because it affects market conditions by shaping consumer demand and the conditions of competition for products. The harder question concerns direct producer subsidies. This section proceeds in three steps. First, it shows that measures directed at producers can still

Distribution of Bananas, ¶ 220, WTO Doc. WT/DS27/AB/R (adopted Sept. 25, 1997).

¹²⁴ Appellate Body Report, *Korea–Measures Affecting Imports of Fresh, Chilled and Frozen Beef*, ¶¶ 142-151, WTO Doc. WT/DS161/AB/R, WT/DS169/AB/R (adopted Jan. 10, 2001); Panel Report, *Turkey–Measures Affecting the Importation of Rice*, ¶ 7.225, WTO Doc. ST/DS334/R (adopted Oct. 22, 2007).

¹²⁵ Appellate Body Report, *Thailand–Customs and Fiscal Measures on Cigarettes from the Philippines*, ¶¶ 129, 134, WT/DS371/AB/R (adopted June 17, 2011) [hereinafter *Thailand–Cigarettes (Philippines)* Appellate Body Report].

fall within the broad coverage of Articles I and III. Second, it explains the limited carve out in Article III:8(b), including the conditionality boundary and its rationale. Third, it returns to the applicable non-discrimination tests for producer subsidies that fall outside Article III:8(b), and it explains why the producer product linkage problem makes findings of discrimination more difficult.

i. Producer Subsidies Are Possibly Within the Broad Coverage

Two cases are relevant for our evaluation of subsidies being subject to non-discrimination obligations. The first is the Panel report in the DSU Article 21.5 proceeding in the *U.S.–FSC* case. In that dispute, the European Communities challenged the U.S.’s Extraterritorial Income Exclusion Act under Article III:4 of the GATT.¹²⁶ The European Communities alleged the U.S.’s exception of foreign income to be calculated as taxable if less than 50 percent of the value of the exported product was produced domestically or using domestic labor fell within the scope of Article III:4.¹²⁷ The U.S. argued with recourse to the drafting history, claiming that incomes tax (or direct taxes) were not subject to the non-discrimination-principal scrutiny.¹²⁸ The Panel reasoned that the drafting history noting Peru’s withdraw that “neither income taxes nor import duties fall within the scope of Article 18 which is concerned solely with internal taxes on goods” was targeting Article III:2 rather than Article III:4.¹²⁹ Additionally, exempting income tax “would entirely exclude an area that was explicitly covered ... Article III:4 of the GATT: i.e., internal regulation.”¹³⁰

The second case *Brazil–Taxation* followed the same reasoning of *U.S.–FSC*. This case concerned a different scenario of preferential indirect taxation, collected whenever an industrial product was undergoing transaction. Brazil argued that their IPI tax (Tax on Industrialized Products) was collected from the producer by the time of leaving an industrial facility rendering it “pre-market,” while Article III were “applicable only to measures which affect a product once it has been produced and enters the marketplace.”¹³¹

¹²⁶ Panel Report, *United States–Tax Treatment for “Foreign Sales Corporations”*, ¶ 3.1(e), Article 21.5 Proceeding, WTO Doc. WT/DS108/AB/RW (circulated Jan. 29, 2002) [hereinafter *U.S.–FSC*]. While the case was appealed, the relevant holding on Article III:4 was affirmed and the drafting history part was not appealed.

¹²⁷ *Id.*, ¶¶ 2.7, 8.123.

¹²⁸ *Id.*, ¶ 8.145.

¹²⁹ *Id.*

¹³⁰ *Id.*

¹³¹ Appellate Body Report, *Brazil–Certain Measures Concerning Taxation and Charges*, ¶ 5.14, WTO Doc. WT/DS472/AB, WT/DS497/AB (adopted Jan. 11, 2019) [hereinafter *Brazil–Taxation* Appellate Body Report].

Both the Panel and Appellate Body refuted this claim.¹³² The Appellate Body claimed that Article III:1 “does not exclude from its scope measures that are on their face directed at producers, which nevertheless subject the product concerned to taxation in excess, and thereby have an impact on the conditions of competition.”¹³³ Article III:4 likewise has wide application, and “measures directed at producers that may affect the ‘specific transactions, activities and uses’ mentioned in Article III:4. (citation omitted).”¹³⁴ However, the Appellate Body was cautious that whether the taxation on foreign product was excessive or the measure treated foreign products less favorable were “determined on the basis of the facts and circumstances at issue.”¹³⁵

Income tax exemptions and foregone revenue can be understood as a common subsidy modality. Treating such measures as subject to non-discrimination disciplines could appear to narrow the SCM Agreement’s role as a distinct framework for subsidy regulation.¹³⁶ However, the Panel’s over-generalized argument in *U.S.–FSC* cannot be accepted too hastily. The Extraterritorial Income Exclusion Act did not chiefly distort competition among final products produced by beneficiaries of the income-tax exclusion. The relevant competitive effect instead arose in input markets, because the scheme conditioned the exclusion on the “foreign articles/labor limitation.” The Panel’s inquiry, accordingly, focused on whether that limitation, rather than the exclusion as such, accorded less favorable treatment to upstream inputs under national treatment. Similarly in *Brazil–Taxation*, the contested tax advantages were conditioned on accreditation requirements tied to the use of domestic inputs, which made a finding of WTO inconsistency comparatively straightforward.

ii. Narrow Carve-Out Under Article III:8(b) Subsidy Exception

While there is a broad coverage of the non-discrimination obligation under the GATT, an exception is added under Article III:8(b), stipulating that “[t]he provisions of this Article shall not prevent the payment of subsidies exclusively to domestic producers, including payments to domestic producers derived from the proceeds of internal taxes or charges

¹³² *Id.*, ¶ 5.16; Panel Report, *Brazil–Certain Measures Concerning Taxation and Charges*, ¶¶ 7.65-71, WTO Doc. WT/DS472/R, WT/DS497/R (circulated Aug. 30, 2017) [hereinafter *Brazil–Taxation Panel Report*].

¹³³ *Brazil–Taxation Appellate Body Report*, ¶ 5.15.

¹³⁴ *Id.*, ¶ 5.16.

¹³⁵ *Id.*, note 119.

¹³⁶ See discussion on the treatment of tax issues under WTO law: Slemrod & Avi-Yonah, *supra* note 118, at 535-37; TURKI ALTHUNAYAN, DEALING WITH THE FRAGMENTED INTERNATIONAL LEGAL ENVIRONMENT: WTO, INTERNATIONAL TAX AND INTERNAL TAX REGULATIONS 153-57 (2010).

applied consistently with the provisions of this Article and subsidies effected through governmental purchases of domestic products.”¹³⁷ Clarifying the scope of this exception serves two purposes. It identifies which measures are excluded from the non-discrimination disciplines, and it also helps explain why measures outside the carve out remain, and should remain, subject to scrutiny under Articles I and III.

The application of this Article is straightforward. The pre-WTO GATT and WTO dispute settlement practices establish a coherent textualism interpretation emphasizing “payment” and “exclusively to domestic producers.” The first dispute concerned Italian’s credit to small farmers for the purchase of domestically produced agricultural machinery over imported alternatives in 1958.¹³⁸ In the *Italy–Agricultural Machinery* case, the Panel acknowledged that, although states are permitted to adopt protectionist measures, such measures must conform to the explicit exceptions outlined in the treaty. The Panel found that the subsidies at issue were granted to consumers rather than to domestic producers. Since the measure did not fall within that exception, Italy violated Article III:4 of the GATT. The Panel clarified that the Italian government is not required to increase its budgetary appropriations, but rather “to extend the availability of the credit facility to permit a fair choice between purchases of tractors of domestic and foreign origin.”¹³⁹

Italy–Agricultural Machinery began a coherently narrow construction of the Article III:8(b) exception. Among the other seven disputes in which Article III:8(b) was invoked, the invocation succeeded in only two cases. In *U.S.–Tobacco*, the Panel accepted that the No-Net-Cost Assessments imposed on imported tobacco constituted price support for domestic producers, and therefore fell within the scope of Article III:8(b).¹⁴⁰ The Panel emphasized that the distinction in Article III:8(b) is a “formal one,” not based on the economic impact of a measure.¹⁴¹ A second successful invocation was in *EC–Vessels*, where the Panel upheld the EU’s state aid program that permitted member states to provide direct financial assistance for domestic shipbuilding contracts.¹⁴² The Panel also addressed the referential clause in Article I:1, holding that the Article III:8(b) exception applies equally to the MFN obligation under Article I.¹⁴³

¹³⁷ GATT art. III:8(b).

¹³⁸ *Italian Discrimination Against Imported Agricultural Machinery* (Oct. 23, 1958) GATT B.I.S.D. (7th Supp.) at 23 (1958) [hereinafter *Italy–Agricultural Machinery*].

¹³⁹ *Id.* ¶¶ 14, 16, 24.

¹⁴⁰ Report of the Panel, *United States–Measures Affecting the Importation, Internal Sale and Use of Tobacco* (Oct. 4, 1994), GATT B.I.S.D. (41st Supp.) at 131, ¶ 111 (1994).

¹⁴¹ *Id.* ¶ 109.

¹⁴² *EC–Vessels* Panel Report, ¶ 4.53, 7.75.

¹⁴³ *Id.* ¶¶ 7.79–7.91.

The invocation of Article III:8(b) failed in *EEC–Oilseeds*, where the payment was made to processors rather than to original producers.¹⁴⁴ The other four disputes, *U.S.–Malt Beverage*,¹⁴⁵ *Canada–Periodicals*,¹⁴⁶ *Indonesia–Autos*¹⁴⁷ and *Brazil–Taxation*¹⁴⁸ all focused on the interpretation of the term “payment.” The measures involved discriminatory tax regimes between domestic and foreign products. The Appellate Body interpreted the term “payment” as having a narrower scope than “subsidy” or “granting or maintaining a subsidy” used elsewhere in the GATT and covered agreements, and it specifically requires an “expenditure of revenue by a government.”¹⁴⁹ Moreover, discriminatory tax regimes on products, though possibly qualifying as subsidies, remain subject to Article III:2 of the GATT.

Following this chain of cases, it is readily clear that only if a subsidy is provided in the form of “payment” and its recipient is limited to domestic producers, will it be permitted to derogate from the non-discrimination obligations. Several lingering questions remains. First, as acknowledged by the *U.S.–Tobacco* Panel, Article III:8(b) is not economically justified since subsidies provided to domestic producer have a similar impact on foreign exporters to that of subsidies contingent upon domestic content, let along the formality requirement of “payment.”¹⁵⁰ Why was the exception drafted in this way and what are its implications? Second, does the exception permit any kinds of conditionality? For instance, would the measure be exempted from MFN scrutiny if the government payment to domestic producer is restricted to those investing in certain state instead of others, or if the contingency is based on input from only certain friend-shores?

For the first question, our inquiry on the drafting history shows that the specific scope of Article III:8(b) possibly derives from a drafting negligence failing to consider treaty systematic coherency.¹⁵¹ The Article was initially negotiated as part of the broader

¹⁴⁴ Report of the Panel, *European Economic Community–Payments and Subsidies Paid to Processors and Producers of Oilseeds and Related Animal-Feed Proteins* (Jan. 25, 1990), GATT B.I.S.D. (37th Supp.) at 86, ¶¶ 138-141 (1990) [hereinafter *EEC–Oilseeds* Panel Report].

¹⁴⁵ Report of the Panel, *United States–Measures Affecting Alcoholic and Malt Beverages* (June 19, 1992), GATT B.I.S.D. (39th Supp.) at 206, ¶¶ 5.8-12 (1992) [hereinafter *U.S.–Malt Beverages* Panel Report].

¹⁴⁶ Appellate Body Report, *Canada–Certain Measures Concerning Periodicals*, 32-35, WTO Doc. WT/DS31/AB/R (adopted July 30, 1997).

¹⁴⁷ Panel Report, *Indonesia–Certain Measures Affecting the Automobile Industry*, ¶¶ 14.118-122, WTO Doc. WT/DS54/R, WT/DS55/R, WT/DS59/R, WT/DS64/R (adopted July 23, 1998).

¹⁴⁸ *Brazil–Taxation* Appellate Body Report, ¶¶ 5.86-91.

¹⁴⁹ *U.S.–Malt Beverages* Panel Report, ¶ 5.11.

¹⁵⁰ SYKES, *supra* note 64, 396.

¹⁵¹ The common notion of constructive ambiguity does not apply here since the article has

discussions on subsidies. In 1946, the First Session of negotiation focused on a draft proposed by the U.S., which was revised into what became known as the London Draft.¹⁵² During the session, the discussion regarding payments of subsidies to domestic producers arose not from a debate on non-discrimination obligations but instead derived from the distinction between export and domestic subsidies.¹⁵³ The negotiators were troubled with taking a formalism prohibition of export subsidies, only prohibiting any subsidies formally granted on exportation, or extending the prohibition to any domestic subsidies that might generate a two-price system. Following the formalism distinction, Australian delegate Edwin McCarthy noted that a subsidy funded through a consumption tax could qualify as a producer subsidy even if it lifted domestic prices above export prices, such as an excise tax raised domestically and paid out to producers. McCarthy maintained that such subsidies shall be permitted rather than being regulated as export subsidies.¹⁵⁴ This was the first documented discussion of the precursor to the GATT Article III:8(b). Two days later, when the meeting reconvened, the US representative Robert Schwenger proposed a clarification that “[t]he use of the proceeds of such duties or taxes to make payments to domestic producers would be considered as a case under paragraph (1).” Under the clarification, “payments to domestic producers” were not export subsidies and prohibited as such, but might be subjected to countervailing measures if injury is caused to other states.¹⁵⁵

Though the clarification was explicitly denounced by the Canadian and Cuban representatives,¹⁵⁶ it remained in the final text of the London Draft.¹⁵⁷ Afterwards, the New York Draft made minor modifications to the London Draft following the US’s

been specific enough. See Andrew T. Guzman, *The Design of International Agreements*, 16 EU. J. INT’L L. 579 (579); Katerina Linos & Tom Pegram, *The Language of Compromise in International Agreements*, 70 INT’L ORG., 587 (2016).

¹⁵² DOUGLAS A. IRWIN, PETROS C. MAVROIDIS & ALAN O. SYKES, *THE GENESIS OF THE GATT*, 98-100 (2008).

¹⁵³ United Nations Economic and Social Council Preparatory Committee of the United Nations Conference on Trade and Employment [hereinafter UNESC Preparatory Committee], *Verbatim Report of the First Meeting of the Joint Drafting Sub-Committee of Committee II and Committee IV on Subsidies on Primary Products* 24, E/PC/T/C.II&IV/PP/PV/1 (Nov. 13, 1946).

¹⁵⁴ McCarthy’s claim explained why Article III:8(b) comes with a very specific example of “payments to domestic producers derived from the proceeds of internal taxes or charges.”

¹⁵⁵ UNESC Preparatory Committee, *Verbatim Report of the Second Meeting of the Joint Drafting Sub-Committee of Committee II And Committee IV on Subsidies on Primary Products* 9, E/PC/T/C.II&IV/PP/PV/2 (Nov. 15, 1946); UNESC Preparatory Committee, *Report of the First Session of the Preparatory Committee of the United Nations Conference on Trade and Employment* 32, E/PC/T/33 (1946).

¹⁵⁶ E/PC/T/C.II&IV/PP/PV/2, *supra* note 155, at 9, 11.

¹⁵⁷ E/PC/T/33, *supra* note 155, at 16.

proposal, and it was agreed upon easily.¹⁵⁸ However, it is noteworthy that the New York Draft expanded the scope of covered subsidies. The definitional scope of subsidies in the London Draft only included those given “to the domestic producers of any product.” Such beneficiary and measure limitation was removed so that “any subsidy” will be covered following the US’s proposal.¹⁵⁹

During the Geneva negotiations, the US proposed inserting an exception into the NT provision.¹⁶⁰ The amendment stated that “the provisions of this Article shall not apply to Government policies in carrying out any form of subsidy permitted under Article 30.” The US’s proposal was considered too broad and incompatible with the NT provisions, since taxes could also be considered as permitted domestic subsidies under the article. As a result, it was referred to an *ad hoc* Sub-Committee on Articles 14, 15, and 24, which modified the proposal to its final version: “nor shall they prevent the payment to domestic producers only of subsidies provided for under Article 25, including payments to domestic producers derived from the proceeds of internal taxes or charges ...”¹⁶¹ The Sub-Committee noted that this revision was intended to align the paragraph with the subsidy provisions in Article 30.¹⁶² It was later explained that this clarification was necessary to prevent an interpretation that NT obligations would prohibit subsidies to domestic producers if similar benefits were not extended to imports.¹⁶³ The subsequent “Havana Draft” merely formalized these changes.¹⁶⁴ The negotiating parties did not revisit Article III:8(b) after this stage.

¹⁵⁸ UNESCO, *Report of the Drafting Committee of the Preparatory Committee of the United Nations Conference on Trade and Employment* 25-26, E/PC/T/34 (Mar. 5, 1947) [hereinafter New York Draft]; Drafting Committee of the Preparatory Committee of the United Nations Conference on Trade and Employment, *United States Suggestions Article 30 General Undertaking Regarding Subsidies - Elimination Of Export Subsidies - Exceptions*, E/PC/T/C.6/W.21 (Jan. 27, 1947).

¹⁵⁹ New York Draft, *supra* note 158, at 25.

¹⁶⁰ UNESCO Second Session of the Preparatory Committee of the United Nations Conference on Trade and Employment, *Ad hoc Sub-Committee on Articles 14, 15 and 24, Note By the Secretariat on Articles 15 and 15-A* 8, E/PC/T/W/181 (June 9, 1947).

¹⁶¹ UNESCO Second Session of the Preparatory Committee of the United Nations Conference on Trade and Employment, *(Draft) General Agreement on Tariffs and Trade* 7 E/PC/T/189 (Aug. 30, 1947).

¹⁶² UNESCO Second Session of the Preparatory Committee of the United Nations Conference on Trade and Employment, *Report to Commission A by the Sub-Committee on Articles 14, 15 & 24* 9, E/PC/T/174 (Aug. 15, 1947).

¹⁶³ United Nations Conference on Trade and Employment, *Third Committee: Commercial Policy Sub-Committee A (Articles 16, 17, 18, 19) Notes of Thirty-Third Meeting* 2 E/CONF.2/C.3/A/W.49 (Feb. 10, 1948).

¹⁶⁴ *Id.*; *Havana Charter for an International Trade Organization*, U.N. Doc. E/CONF.2/78 (Mar. 24, 1948).

The historical record suggests that Article III:8(b) may reflect an honest drafting mistake that gradually lost focus and systematic coherence across successive negotiating stages, so that the negotiators' intended balance was not fully carried through into the final text. Early versions were developed in the context of a narrow understanding of subsidies. Though subsidies discussed during the London session consisted of those provided for both primary and manufactured goods, the negotiation focused predominantly on agricultural commodities, including flour, whiskey, and butter,¹⁶⁵ where subsidies were conceived as payments from consumption tax revenue to support domestic food security and price stabilization.¹⁶⁶ This approach was relatively coherent because the covered subsidies were limited to those granted "to the domestic producers." The predecessors of Article III:8(b), however, were not substantially revised after the New York session expanded the coverage of subsidies, nor after later developments distinguished subsidies for primary and non-primary goods. Once the residual agricultural and primary goods logic is taken into account, Article III:8(b) reads as a narrow tolerance for certain domestic producer payments rather than a general permission for producer selectivity.¹⁶⁷ Therefore, the nature of this Article dictates a limited application no more than what was written.

For the second question, the Appellate Body distinguished between two forms of conditionality under Article III:8(b). One concerns eligibility conditions that define the class of qualified domestic producers in the market for the subsidized product; the other concerns additional requirements, including conditions tied to the input market.¹⁶⁸ The Appellate Body treated the first category as permissible and the second as impermissible. This distinction is relevant to subsidies contingent on the use of domestic or designated foreign inputs. Even where a subsidy is structured as a payment made exclusively to domestic producers, input related conditions can still trigger scrutiny under the national treatment and MFN obligations of the GATT.

iii. Qualified Producer-Product Linkage Issue

After establishing the broad reach of the GATT non-discrimination obligations and the narrow construction of the Article III:8(b) exception, the remaining question is how to assess direct producer subsidies under the non-discrimination principle.¹⁶⁹ The central

¹⁶⁵ E/PC/T/C.II&IV/PP/PV/1, *supra* note 153, at 23-26

¹⁶⁶ *Id.* at 27-28.

¹⁶⁷ See generally, Agreement on Agriculture art. III:8(b), Apr. 15, 1994, Marrakesh Agreement Establishing the World Trade Organization, Annex 1A, 1867 U.N.T.S. 410 (1999).

¹⁶⁸ Brazil–Taxation Appellate Body Report, ¶¶ 5.108, 5.112.

¹⁶⁹ It is worth highlighting that one of the Appellate Body member attached a separate

difficulty is that direct producer subsidies affect the competitive conditions for like products only indirectly. This differs from content contingent subsidies and producer subsidies implemented through consumer side measures, where the competitive impact in the product market is more readily observable and generally pervasive. In those settings, discrimination can often be assessed through a relatively linear inquiry into “the design, structure, and expected operation of the measure at issue.”¹⁷⁰

However, for direct producer subsidies discriminating on the nationality of the producer, there exists a problem of what we termed as producer-product linkage issue.¹⁷¹ Recalling the SCM Agreement controls the disparate provision of subsidies through specificity. The SCM Agreement polices specific subsidies given to “an enterprise or industry or group of enterprises or industries” and “set[s] out indicators as to whether the conduct or instruments of the granting authority discriminate or not.”¹⁷² The *U.S.-Softwood Lumber IV* Panel found that specificity has “to be determined at the enterprise or industry level, not at the product level,” and that “a single industry may make a broad range of end products.”¹⁷³ Hence, even if a subsidy measure differentiates between domestic and foreign producers might be deemed specific, discriminatory in the sense of the SCM

opinion to the Article III:8(b) issue in *Brazil–Taxation*. The member argued that subsidy disciplines only prohibit export and import substitution subsidies, and the careful balance set out in the SCM Agreement should not be abrogated by an overly expansive interpretation of the GATT’s non-discrimination provisions. The member interpreted “payment” as including “a form, other than money, that confers value, such as by way of goods or services.” *Brazil–Taxation* Appellate Body Report, ¶¶ 5.129-36. We agree with the separate opinion to the ground that disparate treatment in subsidization could be certain to extent tolerated. However, while the text controls with the support of drafting history, we argue instead of twisting the ordinary meaning of the texts, the balance can be reached if an adequate non-discrimination test is devised to secure such policy space.

¹⁷⁰ *Thailand–Cigarettes (Philippines)* Appellate Body Report, ¶¶ 129, 134.

¹⁷¹ The issue was identified in the context of discussing the national treatment obligation’s coverage over direct tax. Michael Lennard argued that Article III is “directed at measures which directly or indirectly have sufficient *effect* on, or *linkage* to, the product to disturb competitive conditions.” Though no test for sufficiency was advanced, Lennard argued that “the great majority of direct tax measures sufficiently affect products so as to afford products so as to be regulated by Article III.” Michael Lennard, *The GATT 1994 and Direct Taxes: Some National Treatment and Related Issue*, in *THE WTO AND DIRECT TAXATION* 73, 98, 101 (Michael Lang, Judith Herdin & Ines Hofbauer eds., 2005).

¹⁷² Appellate Body Report, *United States–Definitive Anti-Dumping and Countervailing Duties on Certain Products from China*, ¶ 367, WTO Doc. WT/DS379/AB/R (adopted Mar. 11, 2011); Appellate Body Report, *European Communities and Certain Member States–Measures Affecting Trade in Large Civil Aircraft*, ¶ 943, WTO Doc. WT/DS316/AB/R (adopted June 1, 2011) [hereinafter *EC–LCA* Appellate Body Report].

¹⁷³ Panel Report, *United States–Final Countervailing Duty Determination with Respect to Certain Softwood Lumber from Canada*, ¶ 7.120-21, WTO Doc. WT/DS257/R (circulated Aug. 29 2003).

Agreement, it might not be discriminatory in the product market regulated under Articles I and III of the GATT. In order to channel the specificity of producers to discrimination in treatment to products, it is necessary to establish that the benefits conferred at the “enterprise or industry level” could somehow cause differentiated treatment to the “end products.”

The producer-product linkage issue was most evident rendering the competitive relationship test under Articles I and III arduous in *EC–LCA* and the *U.S.–LCA (Second Complaint)*. A significant portion of the reports in *EC–LCA* was devoted to examining how the EU subsidies regime contributed to the financial capacity of Airbus, eventually enabling the development of the final products A350 and A380, which distorted both domestic and world aircraft markets.¹⁷⁴ Similarly, the European Communities challenged the US’s tax regimes, loans, and R&D contracts with Boeing, arguing that these measures expedited the launch and delivery of the B787.¹⁷⁵ For the causation requirement,¹⁷⁶ various causation theories were advanced, including the “product effect,”¹⁷⁷ “technology effect,”¹⁷⁸ “spill-over technology effect,”¹⁷⁹ and “price effect,”¹⁸⁰ serving the purpose of channeling subsidy benefits to the product market competitive relationship. This complex examination cannot be easily ascertained through the single-layered competitive relationship test under the non-discrimination obligation. Therefore, Articles I:1 and III of the GATT could restrict production subsidies only in a twisted manner, unless the object and purpose of the measures are explicit to impact the upstream input markets.

For these discriminatory subsidies, it is no harm to refer back to the SCM Agreement, the *lex specialis* for subsidy discipline for legal test. Article 6.3 of the SCM Agreement provides a workable solution to the producer-product linkage issue. It stipulates that “the effect of the subsidy is to displace or impede the imports of a like product of another Member into the market of the subsidizing Member.”¹⁸¹ Countervailing those subsidies

¹⁷⁴ Panel Report, *EC–LCA*, ¶ 7.1993, WTO Doc. WT/DS316/R (circulated June 30, 2010) [hereinafter *EC–LCA* Panel Report]; *EC–LCA* Appellate Body Report, ¶ 1356.

¹⁷⁵ Appellate Body Report, *United States–Measures Affecting Trade In Large Civil Aircraft (Second Complaint)*, ¶ 1127, WTO Doc. WT/DS353/AB/R (adopted Apr. 11, 2019) [hereinafter *U.S.–LCA (Second Complaint)* Appellate Body Report].

¹⁷⁶ See Matthew Kennedy, *The Adverse Effects of Technological Innovation Under WTO Subsidy Rules*, 19 WORLD TRADE REV. 511 (2020).

¹⁷⁷ *EC–LCA* Panel Report, ¶¶ 7.1879-81.

¹⁷⁸ *U.S.–LCA (Second Complaint)* Appellate Body Report, ¶¶ 923-28.

¹⁷⁹ Appellate Body Report, *EC–LCA*, ¶¶ 5.424-43, DSU Article 21.5 Proceeding, WTO Doc. WT/DS353/AB/RW (adopted May 28, 2018).

¹⁸⁰ *U.S.–LCA (Second Complaint)* Appellate Body Report, ¶¶ 1146-60; *EC–LCA* Panel Report, ¶¶ 7.197-98.

¹⁸¹ SCM Agreement art. 6.3.

causing serious prejudice is necessary since they work effectively as tariffs to substitute foreign for domestic products.¹⁸² From a legal policy perspective, disciplining serious prejudice serves exactly the same purpose as regulating selective producer subsidies, restricting the substitution of certain designated foreign products for domestic or some foreign products. Additionally, applying the serious prejudice test comes with two heightened requirements: first, more statistical and empirical evidence of prejudice; second, a more reliable causation link between the measure and the alleged trade-distortive effects. For the former, prejudice is classified as displacement and impedance. Displacement must be “clearly identifiable” on the basis of objective evidence, focusing on observable and clear market trends, including both volumes and market shares over a sufficiently representative period.¹⁸³ Impedance encompasses a broader set of circumstances, referring to situations where the imports or exports of a like product from the complaining Member would have expanded but for being “obstructed” or “hindered” by the presence of the subsidized product in the market.¹⁸⁴ Even under a threat of a serious prejudice claim, the threat “is imminent such that it will materialize in the near future,”¹⁸⁵ and should “be based on facts and not merely on allegation, conjecture, or remote possibility.”¹⁸⁶

The heightened evidentiary and causation requirements under the serious prejudice test provide a more appropriate threshold for assessing discriminatory subsidies that directly benefit producers for three reasons. First, the competitive relationship inquiry under GATT Articles I:1 and III:4 is typically single layered and focuses on the immediate effects of a measure on market access and conditions of competition. Because such measures operate directly in the product market, discriminatory effects can often be assessed without extensive empirical proof. Producer subsidies, by contrast, represent an initial intervention that works through subsequent decisions on production, pricing, marketing, and sales. Stronger evidence is therefore needed to demonstrate actual market harm. Second, the more demanding causation element reflects the practical difficulty of showing how benefits conferred on producers translate into competitive advantages for the subsidized products. In the subsidy setting, intervening factors can dilute, offset, or

¹⁸² Alan O. Sykes, *The Limited Economic Case for Subsidies Regulation* 10-11 (John M. Olin L. & Econ., Working Paper No. 472), http://ssrn.com/abstract_id=2531051; Daniel Brou & Michele Ruta, *A Commitment Theory of Subsidy Agreements* 28-29 (World Trade Org., Staff Working Paper ERSD-2012-15), https://www.wto.org/english/res_e/reser_e/ersd201215_e.pdf.

¹⁸³ EC–LCA Appellate Body Report, ¶¶ 1165-66.

¹⁸⁴ *Id.*, ¶ 1161.

¹⁸⁵ Appellate Body Report, *United States–Subsidies on Upland Cotton*, ¶ 244, WTO Doc. WT/DS267/AB/R (adopted Mar. 21, 2005).

¹⁸⁶ EC–LCA Appellate Body Report, ¶ 1171.

independently shape competitive outcomes among like products. Third, the serious prejudice framework provides a coherent approach to the relationship between the GATT and the SCM Agreement. It recognizes Members' discretion to choose subsidy beneficiaries while requiring regimes to be designed and implemented in a manner that limits harm to the product market.

2. *The Outreach of Art. XX and XXI with Regards to Subsidies*

After clarifying the scope of the GATT non-discrimination obligations, the analysis turns to the principal exceptions that Members may invoke to defend resilience related subsidies. For measures aimed at greening supply chains, Members may rely on Article XX(b), which covers measures “necessary to protect human, animal or plant life or health,” or Article XX(g), which applies to measures “relating to the conservation of exhaustible natural resources.”¹⁸⁷ For supply chain governance grounded in human rights or labor concerns, some commentators have argued for an analogous use of Article XX(e), which refers to measures “relating to the products of prison labour.”¹⁸⁸ For subsidies that exclude certain origins based on alleged overcapacity, Article XX(a) on public morals has also been invoked.¹⁸⁹ Members may also invoke the national security exception in Article XXI on the theory that supply chain stabilization and resilience fall within essential security interests, i.e. economic security.¹⁹⁰

¹⁸⁷ Chen & Shen, *supra* note 32, at 16-17.

¹⁸⁸ Aleydis Nissen, *Import Bans on Products from Forced Labor in the Trump Era*, 6 U. BOLOGNA L. REV. 376 (2021); Federico Lenzerini, *International Trade and Child Labour Standards*, in ENVIRONMENT, HUMAN RIGHTS AND INTERNATIONAL TRADE 287, 301-302 (Francesco Francioni ed., 2001).

¹⁸⁹ The public moral exception has been a litigation strategy for the U.S. in various WTO cases, none of which by far have succeeded. See e.g. Panel Report, *United States–Certain Tax Credits Under the Inflation Reduction Act*, ¶ 7.209, WTO Doc. WT/DS623/R (circulated Jan. 30, 2026) [hereinafter U.S.–IRA Panel Report]; Panel Report, *United States–Tariff Measures on Certain Goods from China*, ¶ 7.242, WTO Doc. WT/DS543/R (circulated Sep. 15, 2020).

¹⁹⁰ There is an expounding literature discussing the law and policy of the national security exception under the GATT. Economic security is a long-lasting topic that is mushy conceptionally which might have been clear domestically but is generating significant challenges to the modern trade law regime. Older scholarship grasped national security from a more limited perspective, calling for prevention of abuse and the legalization of the exception, see Michael J. Hahn, *Vital Interests and the Law of GATT: An Analysis of GATT's Security Exception*, 12 MICH. J. INT'L L. 558 (1991); Raj Bhala, *National Security and International Trade Law: What the GATT Says, and What the United States Does*, 19 U. PA. J. INT'L ECON. L. 263 (1998); Hannes L. Schloemann & Stefan Ohlhoff, “Constitutionalization” and Dispute Settlement in the WTO: National Security as an Issue of Competence, 93 AM. J. INT'L L. 424 (1999). With the coming of the Trump administration and flooding cases invoking national security, scholarship starts to tackle

These justifications are attractive because they can expand Members' policy space when subsidies are used as supply chain policy instruments. However, subsidies are regulated under both the GATT and the SCM Agreement, two doctrinal questions shall be answered. The first is whether Articles XX and XXI are available to justify subsidies at all, particularly where the challenged measure falls within the scope of the SCM Agreement. The second is how, if available, these exceptions would operate to shield the subsidizing Member from scrutiny under the GATT non-discrimination obligations or the SCM Agreement's subsidy disciplines.

As to the first question, Articles XX and XXI are available to justify measures challenged under the GATT. The harder issue concerns measures disciplined under the SCM Agreement. In practice, a complaint under SCM Agreement Article 3.1(b) concerning domestic content contingency is often paired with a claim under GATT Article III:4.¹⁹¹ This raises the question whether a justification under Articles XX or XXI can operate across both agreements, curing violations under the GATT and the SCM Agreement in the same dispute.¹⁹² A related problem arises when a measure violates only the SCM Agreement. For example, a subsidy may qualify as a payment made exclusively to domestic producers and thus fall within the Article III:8(b) carve out, yet still be actionable under the SCM Agreement. The question then is whether Articles XX or XXI can justify that subsidy notwithstanding its inconsistency with the SCM Agreement.

this issue from a domestic law and policy ground, claiming the binary of trade and economic security co-existed from the outset of the ITO, and addressing abuses shall come from the inside coupling with international negotiation process, *see e.g.* Kathleen Claussen, *Trade's Security Exceptionalism*, 72 STAN. L. REV. 1097 (2020); Timothy Meyer & Ganesh Sitaraman, *Trade and the Separation of Powers*, 107 CALIF. L. REV. 583 (2019); J. Benton Heath, *The New National Security Challenge to the Economic Order*, 129 YALE L.J. 1020 (2020); Mona Paulsen, *The Past, Present, and Potential of Economic Security*, 50 YALE J. INT'L L. 222 (2025).

¹⁹¹ A nuanced test to determine discrimination hinging upon contingency requirement exists under Article 3.1(b) of the SCM Agreement compared to the competitive relationship test under the GATT. Dominic Coppens argued that the Appellate Body is maintaining a higher threshold of violation of Article 3.1(b) of the SCM Agreement than the GATT. COPPENS, *supra* note 118, at 191.

¹⁹² If being realist on this question, if the Article XX exceptions could potentially justify a domestic content requirement (of course the chances seem remote) under Article III:4 of the GATT, it is extremely unlikely the examination under the SCM Agreement would render such contingency violating Article 3(b) of the SCM Agreement. The Panel or Appellate Body could possibly use the similar reasoning in *Canada-FIT* to show that without governmental intervention, the market could not have existed, hence no benefits could be conferred and no subsidies could be granted. Appellate Body Report, *Canada—Certain Measures Affecting the Renewable Energy Generation Sector / Canada—Measures Relating to the Feed-In Tariff Program*, ¶ 5.185, WTO Doc. WT/DS412/AB/R, WT/DS426/AB/R (adopted May 24, 2013).

Scholars have debated whether the GATT exceptions should be incorporated into the SCM Agreement. Luca Rubini summarized the major arguments from both camps.¹⁹³ Proponents, drawing on multiple interpretive approaches, contend that Article XX can apply on a case-by-case basis because its broadly framed, quasi constitutional role within the GATT is compatible with the WTO's single undertaking architecture.¹⁹⁴ They also argue that the lapse of the non-actionable subsidy provisions does not clearly displace Article XX, and that the SCM Agreement contains no decisive textual barrier or express cross reference that forecloses its application.¹⁹⁵ Nu Ri Jung reaches a similar view by emphasizing the General Interpretative Note to Annex 1A and the *lex specialis* relationship between the SCM Agreement and the GATT. On this account, the GATT can apply insofar as it does not conflict with the SCM Agreement, and prohibited subsidies under SCM Agreement Article 3 remain unjustifiable under Article XX.¹⁹⁶ Related teleological arguments likewise support a case-by-case inquiry, particularly in the context of climate change mitigation.¹⁹⁷ Opponents respond that importing Article XX would upset the SCM Agreement's negotiated internal balance, generate system wide spillovers that could erode market access across covered agreements, and lacks a textual basis because Members add express cross references when they intend exceptions to carry over.¹⁹⁸

WTO practice reflects different approaches to the availability of GATT exceptions beyond the immediate context of GATT claims. In *China—Publications and Audiovisual Products*, the Appellate Body permitted reliance on Article XX in relation to China's Accession Protocol, emphasizing paragraph 5.1's recognition of China's right to regulate trade consistently with the WTO Agreement. Consistency was understood to allow a Member to invoke an applicable exception where the underlying obligation otherwise

¹⁹³ Luca Rubini, *Ain't Wastin' Time No More: Subsidies for Renewable Energy, The SCM Agreement, Policy Space, and Law Reform*, 15 J. INT'L ECON. L. 525, 558-67 (2012).

¹⁹⁴ *Id.*; Robert Howse, *Climate Change Mitigation Subsidies and the WTO Legal Framework: A Policy Analysis*, INT'L INST. SUSTAINABLE DEV. 17 (May 2010), https://www.iisd.org/system/files/publications/bali_2_copenhagen_subsidies_legal.pdf; Roland Ismer et al., *Supporting the Transition to Climate-Neutral Production: An Evaluation Under the Agreement on Subsidies and Countervailing Measures*, 26 J. INT'L ECON. L. 224-25 (2023).

¹⁹⁵ Rubini, *supra* note 193, at 563.

¹⁹⁶ Nu Ri Jung, *Are There "Exceptions" to the SCM Agreement? Applicability of the GATT Exceptions Vis-à-Vis the International Rules on Subsidies*, 57 J. WORLD TRADE 457 (2023).

¹⁹⁷ See e.g. James Bacchus, *The Content of a WTO Climate Waiver* (Ctr Int'l Gov. Innovation Paper No. 204, Dec. 2018), <https://www.cigionline.org/sites/default/files/documents/Paper%20no.204web.pdf>.

¹⁹⁸ Rubini, *supra* note 193, at 562; Ilaria Espa & Gracia Marín Durán, *Renewable Energy Subsidies and WTO Law: Time to Rethink the Case for Reform Beyond Canada—Renewable Energy/FIT Program*, 21 J. INT'L ECON. L. 621 (2018).

would be breached.¹⁹⁹ However, in *China–Raw Material*, both the Panel and the Appellate Body maintained that “were GATT Article XX intended to apply to Paragraph 11.3 of China’s Accession Protocol, language would have been inserted to suggest this relationship.”²⁰⁰ *China–Raw Material* dictates a textual linkage between the justifying clause and the GATT exceptions is necessary.

Two further disputes illustrate how this question arises in subsidy litigation. In *Brazil–Taxation*, Brazil’s domestic content requirement was deemed violating the GATT, the TRIMs (which makes direct reference to the Article XX exception), and the SCM Agreement. However, Brazil only invoked Article XX to justify the GATT and the TRIMs violations.²⁰¹ In the recent *U.S.–IRA*, the U.S. invoked Articles XXI and XX(a) as defenses for certain tax credit measures.²⁰² China later withdrew its claims concerning the Clean Vehicle Credit after the program was terminated by the One Big Beautiful Bill Act.²⁰³ With that claim removed, the panel proceeded on the remaining measures, and the applicability of Article XX to the SCM Agreement was not contested by China. The panel therefore applied the familiar two-step framework associated with Article XX to the claims before it, including those addressing domestic content requirements under the GATT, the TRIMs Agreement, and the SCM Agreement.

It is notable that the question whether GATT exceptions apply to the SCM Agreement has remained unresolved over two decades of WTO dispute settlement. One explanation is strategic indeterminacy. The same features that give repeated judicial reasoning its normative force can also create risks for members that prefer to avoid crystallizing a doctrine through adjudication. Harlan Grant Cohen described three dynamics through which precedent acquires authority: rationalist incentives to invoke prior decisions to advance objectives and stabilize regimes, the jurisprudential pull of the rule of law that makes prior interpretations harder to disregard as they are cited, and sociological

¹⁹⁹ Appellate Body Report, *China–Measures Affecting Trading Rights and Distribution Services for Certain Publications and Audiovisual Entertainment Products*, ¶ 223, WTO Doc. WT/DS363/AB/R (adopted Jan. 19, 2010).

²⁰⁰ Panel Report, *China–Measures Related to the Exportation of Various Raw Materials*, ¶ 7.153, WTO Doc. WT/DS394/R, WT/DS395/R, WT/DS398/R (circulated July 5, 2011); Appellate Body Report, *China–Measures Related to the Exportation of Various Raw Materials*, ¶ 303, WTO Doc. WT/DS394/AB/R, WT/DS395/AB/R, WT/DS398/AB/R (adopted Feb. 22, 2012).

²⁰¹ Brazil–Taxation Panel Report, ¶ 7.850.

²⁰² First Written Submission of the United States of America, *U.S.–IRA*, 17, 20 (Mar. 21, 2025).

²⁰³ Additional Comments of the United States of America to China’s Comments on U.S. Responses to the Panel’s Questions Following the Second Substantive Meeting of the Panel with the Parties, *U.S.–IRA*, 1 (Oct. 10, 2025).

reinforcement through professional communities that treat certain sources as authoritative and reward precedent-based argument.²⁰⁴ In *China–Publications and Audiovisual Products* and *China–Raw Material*, the contested legal instrument was China’s Accession Protocol, so a ruling on exceptions could be framed as limited to a transitional, Member-specific instrument. The stakes are higher when a justification could reshape the disciplines on subsidies and countervailing measures more broadly. Members that may benefit from an exceptions to subsidies including those whose industrial policies rely on domestic content conditions face the risk that an adverse holding would be invoked repeatedly in later disputes. Once embedded in WTO jurisprudence, an unfavorable doctrine can be difficult to dislodge.²⁰⁵

For now, the analysis has largely remained focused on less contentious cases with a sufficiently clear policy foreground. Environmental protection could be the most persuasive basis for deviating from trade discipline. In interpreting the SCM Agreement, it also seems important that environmental obligations derived from major multilateral environmental agreements carry interpretive value. Supply chain resilience, however, is more contentious within WTO disciplines. The line between resilience and protectionism is often difficult to discern, especially in the case of domestic content contingency. Resilience may be an imperative need in some settings, yet it can also be pursued through less trade distortive designs. The economic studies discussed earlier are similarly cautious and point to the need for intricate policy design. From a policy perspective, reliance on the Article XXI security exception may be easier where incentives target critical goods with military or dual use functions. In other settings, justifications framed as securing industrial leads or strengthening potential supply chains against possible obstruction may offer a weaker policy ground.

Assuming the general exceptions are applicable, they may still be difficult to invoke successfully. The Appellate Body has established a two-step test for justification under Article XX.²⁰⁶ First, the measure must fall within at least one of the listed objectives. Second, the measure must satisfy the *chapeau* requirements. States might invoke public morals, protection of human, animal and plant life or health, or conservation of exhaustible natural resources in different contexts, yet those arguments fall beyond the scope of this

²⁰⁴ Harlan Grant Cohen, *Theorizing Precedent in International Law*, in INTERPRETATION IN INTERNATIONAL LAW 268 (Andrea Bianchi et al. eds., 2015).

²⁰⁵ Karen J. Alter, *Resolving or Exacerbating Disputes? The WTO’s New Dispute Resolution System*, 79 INT’L AFF. 783, 795 (2003).

²⁰⁶ Appellate Body Report, *United States–Import Prohibition of Certain Shrimp and Shrimp Products*, ¶¶ 115-21, WTO Doc. WT/DS58/AB/R (adopted Nov. 6, 1998).

discussion.²⁰⁷ This section is limited to what appears to be the most straightforward provision for supply related measures, Article XX(j), which provides that nothing in the GATT 1994 shall be construed to prevent the adoption or enforcement by any Member of measures “essential to the acquisition or distribution of products in general or local short supply,” provided that such measures are consistent with the principle that all Members are entitled to an equitable share of the international supply of such products, and that any measures inconsistent with other provisions of the Agreement are discontinued as soon as the conditions giving rise to them cease to exist.²⁰⁸

Two cases applied this exception in the context of energy security. In *India–Solar Cells*, India’s Jawaharlal Nehru National Solar Mission conditioned eligibility for certain government-linked power purchase arrangements on solar power developers using India-made solar cells and/or modules rather than imported ones. The measures were found violating Article III:4 of the GATT and Article 2.1 of the TRIMs Agreement.²⁰⁹ In *EU–Energy Package*, Russia challenged EU’s TEN-E (Trans-European Networks–Energy) Regulation: its criteria and incentives for designating “Projects of Common Interest,” including a “diversification of gas supply” criterion, were found to provide more favorable conditions for transporting non-Russian gas and thus to breach Articles I:1 and III:4 of the GATT.²¹⁰

Both India and EU’s invocations were unsuccessful. First, “short supply” is an objective market condition. In *India – Solar Cells*, both the Panel and the Appellate Body rejected the idea that a Member’s lack of domestic manufacturing capacity itself constitutes shortage, and instead framed the inquiry as whether the quantity available from both domestic and international sources are insufficient to meet demand in the relevant market, with attention to factors such as market definition, production and import availability, export patterns, price dynamics, purchasing power, and the stability of international supply.²¹¹ Second, the shortage must be temporally concrete rather than speculative. The

²⁰⁷ See *supra* notes 187-189 and accompanying texts.

²⁰⁸ GATT art. XX(j). For some comments on this exception, see Siyu Bao, *Public Policy Defences in the Name of Security: Unconventional Security Claims in WTO General Exceptions Jurisprudence*, WORLD TRADE REV. 1 (2026).

²⁰⁹ Panel Report, *India–Certain Measures Relating to Solar Cells and Solar Modules*, ¶ 7.187, WTO Doc. WT/DS456/R (circulated Feb. 22, 2016) [hereinafter *India–Solar Cells Panel Report*].

²¹⁰ Panel Report, *European Union and its Member States–Certain Measures Relating to the Energy Sector*, ¶¶ 7.1314-24, WTO Doc. WT/DS476/R (circulated Aug. 10, 2018) [hereinafter *EU–Energy Package Panel Report*].

²¹¹ *India–Solar Cells Panel Report*, ¶¶ 7.228-29; Appellate Body Report, *India–Solar Cells*, ¶¶ 5.69, 5.71, WTO Doc. WT/DS456/AB/R (adopted Oct. 14, 2016) [hereinafter *India–Solar Cells Appellate Body Report*].

Panel in *India–Solar Cells* emphasized that Article XX(j) addresses products presently in short supply and, at most, could extend to an imminent shortage,²¹² while the *EU–Energy Package* Panel likewise found that the EU’s reliance on “genuine and serious risks” of disruption was insufficient because Article XX(j) does not cover products that are not currently in short supply but may become shorted later.²¹³ Third, the measure shall be designated to the acquisition or distribution of the product in shortage. On an *arguendo* basis, the *EU–Energy Package* Panel explicitly ruled that EU relied on diversification to reduce disruption risks. It focused on the potential causes of future shortage, not measures that directly secure acquisition or distribution in a shortage.²¹⁴

The reports applying Article XX(j) significantly limit WTO Members’ recourse to justify supply chain measures through a seemingly well-fitting exception. The narrow interpretation of this subparagraph already shows how difficult it is to rely on Article XX(j), particularly where supply chain measures are adopted preventively through friend-shoring or reshoring. This reading is particularly unfavorable to supply chain resilience measures through subsidization, because subsidies take a longer period of time to take effect and only later contribute to resilience. If Article XX(j) is largely inoperable in this setting, resilience subsidies are left to the mercy of other exceptions, which offer even more limited justificatory force.

As for the national security exception, Article XXI(b) is pertinent, prescribing that “Nothing in this Agreement shall be construed (b) to prevent any contracting party from taking any action which it considers necessary for the protection of its essential security interests (i) relating to fissionable materials or the materials from which they are derived; (ii) relating to the traffic in arms, ammunition and implements of war and to such traffic in other goods and materials as is carried on directly or indirectly for the purpose of supplying a military establishment; (iii) taken in time of war or other emergency in international relations.” Subparagraph (iii) is particularly vulnerable to abuse given the breadth of the phrase “emergency in international relations.”

The *Russia–Traffic in Transit* Panel articulated limits on this clause, ruling that the self-judging part “it considers necessary” attaches to the assessment of necessity, and does not permit Members to determine unilaterally whether the circumstances described in subparagraph (iii) exist.²¹⁵ On the meaning of “emergency in international relations,” the

²¹² *India–Solar Cells* Panel Report, ¶¶ 7.263-64 (the holding was not challenged or repealed during appeal).

²¹³ *EU–Energy Package* Panel Report, ¶¶ 7.1348-51.

²¹⁴ *Id.*, ¶ 7.1365.

²¹⁵ Panel Report, *Russia–Measures Concerning Traffic in Transit*, ¶¶ 7.65-66, WTO Doc.

Panel emphasized its proximity to “war” and limited it to “situations of armed conflict, or of latent armed conflict, or of heightened tension or crisis, or of general instability engulfing or surrounding a state.”²¹⁶ The *Saudi Arabia–IP right Protection* Panel extended “emergency” to non-military scenarios, including “grave and serious ... deterioration and rupture in relations between these Members.”²¹⁷ Additionally, the self-judging necessity test shall also be subject to good-faith judicial scrutiny. A state shall “articulate the interest sufficient enough to demonstrate their veracity” and there satisfy the “minimum requirement of plausibility” that the measures would propel the proffered interests.²¹⁸ A supply chain resilience subsidy may be able to satisfy this good faith plausibility standard, yet the main difficulty is establishing that the measure was taken “in time of war or other emergency in international relations.” Under current WTO jurisprudence, domestic supply shortages alone are unlikely to qualify as “latent armed conflict” or as a near rupture of international relations between Members. A more viable argument may arise under Article XXI(b)(ii) where the measures concern military goods or inputs for a military establishment.

3. *Free-Trade Areas Exception Under Art. XXIV*

Several international agreements have been established to strengthen cooperations among states to enhance supply chain resilience. Under the Indo Pacific Economic Framework for Prosperity supply chain agreement, the parties created standing bodies, including the Supply Chain Council and the Supply Chain Crisis Response Network, to identify vulnerabilities and coordinate responses to disruptions.²¹⁹ Alongside these horizontal arrangements, Members have also concluded commodity specific instruments, especially in critical minerals.²²⁰ The U.S.-Japan Critical Minerals Agreement signed in 2023 covers battery-critical minerals (cobalt, graphite, lithium, manganese, nickel) and centers on trade facilitation, market-oriented conditions, and cooperation on labor/environmental standards and supply-chain security.²²¹ The EU’s “raw-materials

WT/DS512/R (adopted Apr. 26, 2019).

²¹⁶ *Id.*, ¶ 7.76.

²¹⁷ Panel Report, *Saudi Arabia–Measures Concerning the Protection of Intellectual Property Rights*, ¶¶ 7.263-64, WTO Doc. WT/DS567/R (circulated June 16, 2020).

²¹⁸ *Id.*, ¶¶ 7.280-81.

²¹⁹ Indo-Pacific Economic Framework for Prosperity arts. 6.1, 6.7(b), 6.13, 7.1, 7.2, 11.1, 12.1.

²²⁰ See generally, Sunayana Sasmal, *Let’s Get Critical: Critical Minerals Mini Deals as Evolving Models of Trade Cooperation*, WORLD TRADE REV. 1 (2026).

²²¹ Agreement Between the Government of the United States of America and the Government of Japan on Strengthening Critical Minerals Supply Chains, U.S.-Japan, Mar. 28, 2023, T.I.A.S. No. 23-328.

diplomacy” has produced a growing network of strategic partnerships with suppliers including Argentina, Australia, Canada, Chile, DRC, Kazakhstan, Namibia, Norway, Serbia, Ukraine, Zambia.²²² It also kept the EU-U.S. Critical Minerals Agreement on the table (negotiations authorized in 2023) to deepen transatlantic supply chains and align sustainability/labor commitments.²²³ Most recently, the U.S., European Commission, and Japan agreed to develop Action Plans featuring coordinated trade tools to mitigate critical minerals supply vulnerabilities. The action plan includes using “price gap subsidies” among others to develop coordinated trade policies.²²⁴

Some regimes provide subsidies directly to preferred trading partners. The repealed IRA, as a subsidy contingent on the use of designated foreign content, linked the critical minerals component of the Section 30D clean vehicle credit to sourcing in the U.S. or in an “FTA partner.”²²⁵ In practice, “FTA partner” was interpreted to include not only comprehensive free trade agreements with broad concessions and disciplines, but also single commodity arrangements, including the U.S.-Japan Critical Minerals Agreement. Similarly, the EU Critical Mineral Act channels foreign production support primarily to producers located in states that have concluded strategic partnerships with the EU. If these measures are found to breach the MFN obligation under the GATT, the key question is whether they can be defended under the Article XXIV free trade area exception.

The case is limited. Article XXIV of the GATT operates as a conditional departure from MFN, and its legal architecture is built around three linked requirements. Article XXIV:4 supplies the object-and-purpose frame, stipulating that regional arrangements are tolerated only insofar as they facilitate trade among the parties without raising barriers to third countries.²²⁶ Article XXIV:8 defined “free-trade area” or “customs union,” requiring

²²² *Raw Materials Diplomacy*, EUR. COMM’N, https://single-market-economy.ec.europa.eu/sectors/raw-materials/areas-specific-interest/raw-materials-diplomacy_en (last visited Feb. 24, 2026).

²²³ *Trade with the United States: Council Authorizes Negotiations on EU-U.S. Critical Minerals Agreement*, COUNCIL EUR. UNION (July 20, 2023), <https://www.consilium.europa.eu/en/press/press-releases/2023/07/20/trade-with-the-united-states-council-authorises-negotiations-on-eu-us-critical-minerals-agreement/>; Council Decision (EU) 2023/1560 of 20 July 2023 Authorizing the Opening of Negotiations with the United States of America for an Agreement on Strengthening Supply Chains for Critical Minerals, 2023 O.J. (L 190).

²²⁴ *Ambassador Jamieson Greer Announces Critical Minerals Cooperation with the European Union and Japan*, OFF. U.S. TRADE REP. (Feb. 4, 2026), <https://ustr.gov/about/policy-offices/press-office/press-releases/2026/february/ambassador-jamieson-greer-announces-critical-minerals-cooperation-european-union-and-japan>.

²²⁵ IRA, § 13401 (as codified in 26 U.S.C. § 30D (2025)).

²²⁶ GATT art. XXIV:4.

the elimination of duties and “other restrictive regulations of commerce” on trade in originating products with respect to “substantially all trade” between the parties.²²⁷ These definitional restrictions functions as the principal anti-abuse constraint, preventing Members from using a purported FTA to justify selective, commodity-by-commodity preferences that would hollow out MFN. Finally, Article XXIV:5 discusses the external relationship between the free trade area or customs union with non-members. Their formation must not result in higher or more restrictive barriers vis-à-vis third countries than existed prior to the agreement.²²⁸

Taken together, these subsections imply that Article XXIV can justify MFN departures only for agreements that entail broad, cross-sector liberalization, rather than narrowly targeted, single-commodity cooperative instruments.²²⁹ Critical-minerals agreements focused on facilitation, standardization, or fulfillment of eligibility for subsidies that do not eliminate duties or “other restrictive regulations of commerce” across substantially all trade between the parties. Even considering the broader range agreements including IPEF, it still does not qualify under the Article XXIV exception without dictating additional and more preferential treatment.

C. Brief Summary

Our review of WTO law and jurisprudence from the GATT period through the WTO period suggests that, although WTO disciplines on non-discrimination may appear unfavorable to resilience-oriented policies, they do not foreclose the use of subsidies as a tool for strengthening supply chain resilience.²³⁰ The central question is how to structure such measures in a manner that advances resilience objectives while remaining consistent with the applicable WTO obligations. Content contingent subsidies remain the most legally exposed category of resilience measures. Subsidies contingent on the use of domestic content face dual vulnerability. They are prohibited under SCM Agreement Article 3.1(b) and they are also susceptible to challenge under GATT Article III:4. Subsidies contingent on designated foreign content face less direct discipline under the SCM Agreement due to the lack of a *per se* prohibition. These measures nonetheless could be challenged under Article I:1 of the GATT as they alter the competitive relationship among foreign like content products. They may also be actionable under the SCM Agreement where a complainant can establish adverse effects, especially serious prejudice through

²²⁷ GATT art. XXIV:8.

²²⁸ GATT art. XXIV:5.

²²⁹ MAVROIDIS, *supra* note 67, at 299.

²³⁰ Meyer, *supra* note 31, at 643.

displacement or impediment.

The analysis becomes more intricate for producer-oriented subsidies, because their effects on the competitive relationship among like products are often indirect. Under the SCM Agreement, even unlikely if the measures are carefully designed, reshoring domestic production subsidies can still be actionable, and serious prejudice may arise if other states' products are displaced or impeded from entering the subsidizing market. Under the GATT, the same measures can raise national treatment concerns when a complainant can establish sufficient producer-product linkage. Though Article III:8(b) seems to secure a policy space for subsidization for subsidies, it is rather limited to subsidies taken the form of a payment made exclusively to domestic producers. Foreign production subsidies raise an additional threshold issue under the SCM Agreement. Friend-shoring measures support production in specified foreign locations with the expectation of importation into the subsidizing Member for consumption or downstream manufacturing. Challenges under the SCM Agreement are complicated by the jurisdictional element embedded in specificity, which ties the subsidy discipline to enterprises or industries within the granting authority's jurisdiction. Apply the MFN obligations would require satisfying the heightened evidential threshold of producer-product linkage.

Finally, the scope for justification under the GATT exceptions appears limited for most resilience subsidies. Article XX(j) is interpreted narrowly, with emphasis on current or imminent short supply, which fits poorly with preventive resilience strategies and with subsidies that require time to take effect. Article XXI likewise imposes a demanding threshold through the requirement of war or other emergency in international relations, and existing jurisprudence makes it difficult to equate domestic shortages with that condition. Article XXIV offers little room for targeted resilience arrangements and product specific critical minerals instruments because they rarely satisfy the "substantially all trade" requirement. Where a measure is challenged under the SCM Agreement, an additional uncertainty remains as to whether and how GATT exceptions can be incorporated into the subsidy discipline. Even if incorporation were accepted, the same doctrinal constraints identified above would still limit its practical value.

V. RECONCEPTUALIZE PERMISSIBLE SUPPLY-CHAIN RESILIENCE SUBSIDIES

Our inquiry in the last section suggests that supply-chain resilience subsidies will almost inevitably introduce some degree of discrimination, whether through eligibility conditions tied to the recipient's nationality or through requirements linked to the origin of

inputs and products. Although the SCM Agreement can constrain certain forms of such measures, its disciplines are often comparatively permissive, and in any event the policy rationale of resilience should not be collapsed into protectionism or the pursuit of additional domestic capacity for export competition. Against this backdrop, the legitimacy of resilience-oriented intervention raises two connected questions. First, what remains of the normative and systemic value of non-discrimination when supply-chain risk and resilience increasingly shape how states structure their external economic relations? Second, how should trade rules evolve so that they preserve meaningful policy space for bona fide resilience measures while still disciplining opportunistic “beggar-thy-neighbor” interventions that merely repackage industrial favoritism as risk management?

A. The Present Value of Non-Discrimination Principle

The non-discriminatory principle serves as one of the “constitutional” rules of international law. The International Law Commission proclaimed that the rule of non-discrimination “is a general rule which follows from the equality of States,”²³¹ and “inherent in the sovereign equality of States.”²³² The principle is essential across different international law regimes, including diplomatic law,²³³ human rights law,²³⁴ and the law of the sea.²³⁵ Discrimination in trade has been commonly viewed as particularly resentful politically, as it provokes tension among countries and manufactures being treated differently, and ultimately poisons international relations.²³⁶ In addition, discrimination distorts the market in favor of products and services that are more expensive and/or of a lower quality.

As for the national treatment, it seeks to prevent internal measures from undermining border commitments, acting as a safeguard against the incompleteness of trade

²³¹ *Document A/3859: Report of the International Law Commission Covering the Work of Its Tenth Session*, 28 April-4 July 1958 [1958] 2 Y.B. Int'l L. Comm'n 105, U.N. Doc. A/CN.4/SER.A/1958/Add. 1.

²³² *Document A/4843: Report of the International Law Commission Covering the Work of Its Thirteenth Session*, 1 May-7 July 1961 [1961] 2 Y.B. Int'l L. Comm'n 128, U.N. Doc. A/CN.4/SER.A/1961/Add. 1.

²³³ EILEEN DENZA, *DIPLOMATIC LAW: COMMENTARY ON THE VIENNA CONVENTION ON DIPLOMATIC RELATIONS* 406 (2016).

²³⁴ Vernon Van Dyke, *Human Rights Without Discrimination*, 67 AM. POL. SCI. REV. 1267 (1973).

²³⁵ United Nations Convention on the Law of the Sea art. 25(3), Dec. 10, 1982, 1833 U.N.T.S. 3.

²³⁶ PETER VAN DEN BOSSCHE, *THE LAW AND POLICY OF THE WORLD TRADE ORGANIZATION: TEXT, CASES AND MATERIALS* 321 (2012).

agreements.²³⁷ Moreover, the national treatment limits protectionist misuse of domestic tools, encouraging greater tariff liberalization and welfare gains.²³⁸ Though several challenges pertain to the utility of offering preferential treatment to domestic, the requirement of providing national treatment to foreign goods or even foreign producers in the case of investment remains untacked, and rule has been one of the most challenged norm under the WTO dispute settlement proceedings. The more contentious case is the remaining values of the MFN obligations.

As early in medieval times, the Holy Roman Emperor promised the medieval city of Mantua that it would always benefit from any privilege granted by the Emperor to “whatsoever other town.”²³⁹ John H. Jackson surveyed that the term “most-favourable-nation” was first utilized at the end of the 17th century.²⁴⁰ MFN obligations was categorized into two types: a conditional MFN extending a concession to the MFN partner only if that partner offers an equivalent concession in return; an unconditional MFN automatically extends any concession granted to any third country to the MFN partner without requiring additional compensation. The modern unconditional MFN proliferated within Europe only after the Cobden-Chevalier Treaty of 1860, reducing tariff and liberalizing trade between Great Britain and France. Within this “mother of all spaghetti bowls,”²⁴¹ an MFN clause guaranteed that a party would not be treated worse than any other state with that either party would have assumed trade relations.²⁴² Ever since, trade agreements network with unconditioned MFN swept across all of the Europe, even if tariff reduction was not included within the agreements.²⁴³

Through empirical studies, Markus Lampe argued three strings of propellants contributed to the sweeping network. First, a welfare economic explanation argued by Kyle Bagwell and Robert W. Staiger theorized that states could raise welfare by an optimal tariff

²³⁷ Henrik Horn, *National Treatment in Trade Agreements*, 96 AM. ECON. REV. 394 (2006); Henrik Horn, Giovanni Maggi & Robert W. Staiger, *Trade Agreements as Endogenously Incomplete Contracts*, 100 AM. ECON. REV. 394 (2010).

²³⁸ Henrik Horn & Petros C. Mavroidis, *Non-Discrimination*, in THE PRINCETON ENCYCLOPEDIA OF THE WORLD ECONOMY 833, 838 (Kenneth A. Reinert et al. eds., 2009).

²³⁹ *Id.*

²⁴⁰ JOHN H. JACKSON, THE WORLD TRADING SYSTEM: LAW AND POLICY OF INTERNATIONAL ECONOMIC RELATIONS 158 (1997).

²⁴¹ Markus Lampe, *Explaining Nineteenth-Century Bilateralism: Economic and Political Determinants of the Cobden–Chevalier Network*, 64 ECON. HIS. REV. 644, 645 (2011).

²⁴² Paul Bairoch, *European Trade Policy, 1815-1914*, in THE CAMBRIDGE ECONOMIC HISTORY OF EUROPE, VOL. 8, THE INDUSTRIAL ECONOMIES: THE DEVELOPMENT OF ECONOMIC AND SOCIAL POLICIES 3 (Peter Mathias & Sidney Pollard eds., 1989).

²⁴³ Douglas A. Irwin, *Multilateral and Bilateral Trade Policies in the World Trading System: An Historical Perspective*, in NEW DIMENSIONS IN REGIONAL INTEGRATION 90, 97 (Jaime de Melo & Arvind Panagariya eds., 1995).

and improve their terms of trade, but states might get retaliation and get caught up in a prisoner's dilemma. Reciprocal tariff cuts can restore a better cooperative outcome and multilateral regime are the first-best option.²⁴⁴ Second, a political economy explanation by Gene M. Grossman and Elhanan Helpman. They argued that governments do not maximize welfare alone, they also maximize political support. Import-competing sectors lobby for tariffs while exporters may favor liberalization. The political equilibrium can therefore generate tariffs and also preferential cooperation that is not globally welfare-maximizing.²⁴⁵ Third, an international interaction theory claiming that the spread of treaties was driven by outsiders' fear of discrimination and exporter lobbying.²⁴⁶

The U.S.'s transition to unconditional MFN was relative later and posited in an economic period of instability. After the First World War, a series of studies by the U.S. Tariff Commission concluded that conditional MFN encouraged foreign governments to discriminate against U.S. trade in order to preserve bargaining leverage for future tariff concessions.²⁴⁷ The Commission argued that unconditional MFN would promote an "open door" policy abroad by offering the attraction of the lowest tariff rates to all countries that adopted non-discriminatory policies. The U.S. Congress's 1922 tariff legislation adjusted the MFN clauses in U.S. trade agreements into an unconditional obligation, though Secretary of State Cordell Hull believed that the bill could pass Congress only because it significantly raised U.S. tariff rates, and that those higher rates would offset whatever liberalizing effect unconditional MFN might otherwise bring.²⁴⁸ If there were no concessions to begin with, there would be no MFN issue in the first place. The unconditional MFN was left in the shadow during the enactment of the notorious Smoot-Hawley Tariff Act, until the Reciprocal Trade Agreements Act of 1934 designed by Hull

²⁴⁴ KYLE BAGWELL & ROBERT W. STAIGER, *THE ECONOMICS OF THE WORLD TRADING SYSTEM* 13-51 (2002).

²⁴⁵ Gene M. Grossman & Elhanan Helpman, *Protection for Sale*, 84 AM. ECON. REV. 833 (1994); Gene M. Grossman & Elhanan Helpman, *The Politics of Free-Trade Agreements*, 85 AM. ECON. REV. 667 (1995).

²⁴⁶ Richard E. Baldwin, *A Domino Theory of Regionalism*, in *EXPANDING MEMBERSHIP OF THE EUROPEAN UNION* 25 (Richard E. Baldwin, Pertti Haaparanta & Jaakko Kiander eds., 1995); David A. Lazer, *The Free Trade Epidemic of the 1860s and Other Outbreaks of Economic Discrimination*, 51 WORLD POL. 51 447 (1999).

²⁴⁷ U.S. TARIFF COMM'N, *SUMMARY OF THE REPORT ON RECIPROCITY AND COMMERCIAL TREATIES WITH CONCLUSIONS AND RECOMMENDATIONS OF THE COMMISSION* 40 (1919); DOUGLAS A. IRWIN, *CLASHING OVER COMMERCE: A HISTORY OF U.S. TRADE POLICY* 363 (2017).

²⁴⁸ Kenneth W. Dam, *Cordell Hull, the Reciprocal Trade Agreements Act, and the WTO*, in *REFORMING THE WORLD TRADE SYSTEM: LEGITIMACY, EFFICIENCY, AND DEMOCRATIC GOVERNANCE* 83, 87 (Ernst Ulrich-Petersmann eds., 2005).

went into force.²⁴⁹ Some argued that Hull may have been fortunate to benefit from the gradual recovery of the world economy in the late 1930s, obtaining renewed negotiating authority in 1937, 1940, and 1943;²⁵⁰ these experiences helped validate the view that reciprocal bargaining and the principle of non-discrimination were consistent with the national interest, and they made MFN a central element of U.S. trade policy.

The following story is familiar to trade lawyers, the GATT was the most successful reflection of the states' consents to the unconditioned MFN obligation.²⁵¹ Economic studies also tried to reveal that multilateral MFN promotes predictability, investment, and multilateral liberalization by preventing discriminatory tariffs and opportunistic policy shifts,²⁵² and helps internalize trade externalities and supports reciprocity in negotiations.²⁵³ Several episodes of resentment for free-riding were prevalent during the Tokyo Round. Industrialized countries complained that smaller countries chose not to participate in reducing trade barriers, yet continued to benefit from major countries' tariff cuts through the MFN principle. In response, the U.S. Congress enacted legislation to apply a conditional MFN approach to the Tokyo Round, and several multilateral agreements concluded during the Round applied only to those GATT contracting parties that signed them.²⁵⁴ However, WTO proceeded on the constitutional principle of unconditional MFN and the MFN clause is among one of the most challenged WTO law provisions. The prohibition of discrimination in trade agreements has shown both concreteness, as distinctively derived not merely from natural morality, and normativity, as distinct from state behaviours, will, and interests.²⁵⁵

Major challenges to the MFN occurred fast-forwarding to the second Trump administration. Reciprocity became the main theme. The Trump administration identifies MFN as causing free riding for those not providing reciprocity to the U.S. In the U.S.'s

²⁴⁹ Douglas A. Irwin, *From Smoot-Hawley to Reciprocal Trade Agreements: Changing the Course of U.S. Trade Policy in the 1930s*, in *THE DEFINING MOMENT: THE GREAT DEPRESSION AND THE AMERICAN ECONOMY IN THE TWENTIETH CENTURY* 325 (Michael D. Bordo, Claudia Goldin & Eugene N. White, eds., 1998).

²⁵⁰ Dam, *supra* note 248, at 87-88.

²⁵¹ Joanne Gowa & Raymond Hicks, *The Most-Favored Nation Rule in Principle and Practice: Discrimination in the GATT*, 7 *REV. INT'L ORG.* 247, 248 (2012).

²⁵² Grossman & Helpman, *supra* note 245; Jacob Viner, *The Most-Favored-Nation Clause in American Commercial Treaties*, 32 *J. POL. ECON.* 101 (1924).

²⁵³ BAGWELL & STAIGER, *supra* note 244, at 92; Wilfred J. Ethier, *Political Externalities, Nondiscrimination, and a Multilateral World*, 12 *REV. INT'L ECON.* 303 (2004).

²⁵⁴ MARTTI KOSKENNIEMI, *FROM APOLOGY TO UTOPIA* 17 (2005).

²⁵⁵ J. Shelton Erb, G.C. Hufbauer & H.P. Starr, *The GATT Codes and the Unconditional Most-Favored-Nation Principle*, 12 *L. & POL'Y INT'L BUS.* 59, 61 (1980); C. Michael Aho, *More Bilateral Trade Agreements Would Be a Blunder: What the New President Should Do*, 22 *CORNELL INT'L L.J.* 25, 28-29 (1989).

WTO reform proposal, the visions embedded in MFN were denounced as “naïve” and blamed lack of “fair, market-oriented competition,” incompatible economic systems, and “pursuit of chronic trade surpluses.”²⁵⁶ The EU’s reform proposal mirrored the U.S.’s emphasis on a step-back from the MFN obligation though in a slightly moderate tone, “[t]his discussion should include reflections on the role of the MFN principle in today’s context, its link to reciprocity and to Members’ respective levels of openness, as well as possible new links between commitments taken and the level of tariffs.”²⁵⁷ These proposals sharply contrasted with that from China, claiming that “blatant violation of the MFN principle ... are eroding the foundation of the rules-based multilateral trading system” and “MFN principle shall remain the bedrock of the WTO.”²⁵⁸

A rationalist view could propose that China supports the unconditioned MFN since it is one of the state’s benefits the most from the MFN tariff reduction and market opening; while the traditional North states are regressing from the principle since the North’s cost of inducing reciprocity is increasing as the South move into export sectors.²⁵⁹ However, several arguments could counter the argument for lack of reciprocity under the current MFN regime.

Jacob Viner’s 1924 account remains workable. Viner emphasized that unconditional MFN should not be understood as a one-shot free ride, but as a continuing reciprocal contract. Looked at “commercial intercourse as a whole,” it produces “as complete and reciprocal an exchange of considerations” as any negotiated bargain. While a particular concession may be obtained in return for compensation at one moment, the MFN pledge executes reciprocity gradually over the life of the treaty by automatically extending each party’s subsequent concessions to the other whenever they are granted to third states. Because each side can expect to share in the partner’s future liberalization without continual renegotiation, unconditional MFN stabilizes bargains, secures expectations, and sustains willingness to exchange market access.²⁶⁰ There is little reason to assume that the underlying logic of the Cobden-Chevalier treaty network that reciprocity through ongoing

²⁵⁶ WTO General Council, *On WTO Reform: Communication from the United States*, WTO Doc. WT/GC/W/984, ¶ 3.1 (Dec. 15, 2025).

²⁵⁷ WTO General Council, *EU Submission on WTO Reform: Communication from the European Union*, WTO Doc. WT/GC/W/986, 3, 8 (Jan. 21, 2026).

²⁵⁸ WTO General Council, *China’s Position Paper on WTO Reform Under the Current Circumstances: Communication from China*, WTO Doc. WT/GC/W/989, ¶¶ 5-6 (Feb. 18, 2026).

²⁵⁹ Pinar Cebi & Rodney Ludema, *The Rise and Fall of the Most-Favored-Nation Clause* (U.S. International Trade Commission Office Of Economics Working Paper No. 2002-06-B), <https://www.usitc.gov/publications/332/ec0206b.pdf>.

²⁶⁰ Viner, *supra* note 252, 118-19, 28-29.

nondiscrimination has disappeared.

On a more general aspect, unconditioned MFN is the “code conditionality,” a prerequisite for joining the WTO framework.²⁶¹ Any MFN treatment granted to a Member is conditioned upon the compliance with all the obligations under the WTO agreements, the violation of which could result in retaliation sometimes across board irrespective of the MFN commitments.²⁶² Additionally, though sitting in the center, reciprocity under the WTO is not merely confined to international trade and tariffs on goods, it has extremely broader coverages. The MFN treatment is a requirement for trade of services and intellectual rights protection.²⁶³ In sectors such as financial services, computer software services, and technology trade, the North continuously extracts profits from technical licensing fees and brand premiums on a global scale, which are made possible only by the unified protection of intellectual property rights.²⁶⁴ In this regard, concessions in services and intellectual property protection can partially offset any perceived reciprocity deficit in trade in goods. This cross-sector balancing is most clearly illustrated by the authorized retaliation against the U.S. in *U.S.–Gambling*, where Antigua and Barbuda was permitted to suspend its TRIPS obligations up to US\$21 million per year.²⁶⁵ Antigua and Barbuda argued that, as a small economy, suspending concessions under the GATS or GATT would be largely ineffective or self-defeating, whereas suspending TRIPS obligations could generate practical leverage against a much larger trading partner.²⁶⁶ Should reciprocity be deficient across the WTO system as a whole, the dispute settlement mechanism and authorized retaliation will remain inoperable for small and developing economies.

Douglas A. Irwin’s observation on the political economy of trade policy is insightful

²⁶¹ JOHN H. JACKSON, *THE WORLD TRADING SYSTEM: LAW AND POLICY OF INTERNATIONAL ECONOMIC RELATIONS* 136-38 (1992).

²⁶² *Id.*

²⁶³ General Agreement on Trade in Services art. II:1, Apr. 15, 1994, Marrakesh Agreement Establishing the World Trade Organization, Annex 1B, 1869 U.N.T.S. 183; Agreement on Trade-Related Aspects of Intellectual Property Rights art. 4, Apr. 15, 1994, Marrakesh Agreement Establishing the World Trade Organization, Annex 1C, 1869 U.N.T.S. 299.

²⁶⁴ Claire Rousseau, Davide Bonaglia & Sacha Wunsch-Vincent, *International Trade in Ideas, Know-how and Intellectual Property Surpasses 1 trillion in 2023; the US and Japan, as Well as Ireland and China Lead*, WORLD INTEL. PROP. ORG. (June 26, 2025), <https://www.wipo.int/en/web/global-innovation-index/w/blogs/2025/international-trade?utm>.

²⁶⁵ Decision by the Arbitrator, *United States–Measures Affecting the Cross-Border Supply of Gambling and Betting Services*, ¶ 6.1, DSU Article 22.6 Arbitration, WTO Doc. WT/DS285/ARB (circulated Dec. 21, 2007).

²⁶⁶ *Id.*, ¶¶ 4.23-24, 4.65; Georgia Hamann, *Replacing Slingshots with Swords: Implications of the Antigua-Gambling 22.6 Panel Report for Developing Countries and the World Trading System*, 42 VAND. J. TRANSNAT’L L. 993, 1018-19 (2009)

and could be transferred here: the benefits of weakening unconditional MFN in goods trade are concentrated among a small set of import-competing producers, while the costs are dispersed across consumers who typically have weaker incentives to organize politically.²⁶⁷ Unconditional MFN under the WTO, however, can generate welfare gains globally and often domestically by institutionalizing the predictability of bargains and sustaining a broader equilibrium in which the North's comparative strengths in services and intellectual property are complemented by the South's role in goods manufacturing. Renouncing MFN's value, especially through turning a blind eye to across-the-board multilateral commitments, is therefore unlikely to deliver durable gains for the North and may instead erode the very conditions that support its long-term advantages.

B. Does Supply Chain Resilience Dictate Discrimination?

Supply chain resilience has become a pervasive policy objective precisely because it reframes interdependence both as an efficiency-enhancing condition of trade and a source of vulnerability. Whereas the non-discrimination principle presupposes that equal competitive opportunity can be maintained by restraining nationality- or origin-based distinctions in governmental measures, resilience policies begin from the opposite intuition: exposure to concentrated dependencies may be intolerable, and the state may need to reengineer the pattern of cross-border sourcing, investment, and production. This conceptual shift creates an immediate tension. The U.S. in its WTO reform document claimed that the WTO forum is not a “productive forum” for dealing with supply chain resilience since the system has a “bias” towards trade liberalization and downplaying risk in the interdependent trading system.²⁶⁸

Yet it would be analytically imprecise to infer from this that resilience dictates discrimination violating the relevant non-discriminatory rules. The relevant question is not whether resilience implies selectivity, but whether selectivity must take the legal form of origin-based preference or exclusion. Resilience-driven selectivity may be implemented through instruments that are explicitly discriminatory: local-content conditions attached to subsidies, procurement preferences for domestic suppliers, eligibility criteria tied to preferred-origin inputs, or friend-shoring schemes that condition market access on geopolitical alignment. These measures most directly collide with the non-discrimination disciplines because they operationalize resilience by formally reallocating competitive advantage along nationality lines. These measures might have a political economy

²⁶⁷ Irwin, *supra* note 247, 20-21.

²⁶⁸ *On WTO Reform: Communication from the United States*, *supra* note 256, ¶ 4.9.

justification domestically, they do not coincident with the current economic findings which requires the heterogeneity of subsidies design, focusing on the product, the specific supply chain and the demand curve rather than a blatant discriminatory rule favoring domestic production or prohibiting the inputs or products from certain states permanently.

In fact, as our analysis of the WTO doctrines show lots of the supply chain resilience measures through subsidization are not easily be deemed as non-compliant. It is worth mentioning the 2023 amendment of the Japan Bank of International Cooperation Act, which broadened JBIC's operational remit to strengthen supply chain resilience and support Japanese firms' global operations we discussed earlier. The amendment proceeds from a risk-oriented scrutiny of funded projects instead of a uniform nationality-based preferences, helping to clarify its non-protectionism objectives. For instance, JBIC provided a loan to Tsumura & Co. to finance production expansion by its Chinese subsidiary so that a consistent production of an intermediate powder for certain Japanese traditional herbal medicine.²⁶⁹ Another is provided to produce printed circuit boards in Jiangsu, China.²⁷⁰ A loan was also provided to Compañía Minera Arqueros S.A. to project a copper mine in Chile.²⁷¹ Another critical mineral loan was provided to a smelting company for production of low-carbon aluminum in Brazil.²⁷² These examples illustrate a diversified subsidization regime that tracks the strategic importance and risk of disruption for different inputs, the heterogeneous demand conditions across sectors, and the manufacturing comparative advantages and industrial ecosystems of host states. Instead of being a single, openly discriminatory design that prioritizes domestic over foreign production or certain designated foreign over other foreign as such, Japanese design of subsidy regime in the form of production loans make them immune from substantial WTO law challenges.

²⁶⁹ *Loan for Kampo Extract Powder Manufacturing Business of Chinese Subsidiary of Tsumura & Co.: Supporting Overseas Business Expansion That Contributes toward Enhancing Supply Chain Resilience of Japanese Pharmaceutical Industry*, JAPAN BANK INT'L COOP. (Aug. 27, 2024), https://www.jbic.go.jp/en/information/press/press-2024/press_00061.html.

²⁷⁰ *Financing for the Manufacturing and Sales of Printed Circuit Boards Carried out by Japanese Companies in China*, JAPAN BANK INT'L COOP. (Mar. 14, 2025), https://www.jbic.go.jp/ja/information/press/press-2024/press_00148.html.

²⁷¹ *Loan for Arqueros Copper Mine Project in Chile: Supporting Japan's Long-term, Stable Supply of Copper Concentrate*, JAPAN BANK INT'L COOP. (Apr. 26, 2024), https://www.jbic.go.jp/en/information/press/press-2024/press_00006.html.

²⁷² *Loan to ALBRAS—Alumínio Brasileiro S.A. in Brazil: Supporting Japanese Aluminum Industry in Securing Long-Term Stable Supply of Low-Carbon Aluminum and Enhancing Supply Chain Resilience*, JAPAN BANK INT'L COOP. (Oct. 29, 2025), https://www.jbic.go.jp/en/information/press/press-2025/press_00103.html.

At the same time, the persistence of the non-discrimination principle does not deny that some resilience concerns may be legitimate and acute especially in the case of global pandemic. The current abuse-reversive jurisprudence of the two exceptions under the GATT in effect erases much, even too much due to strictly textualist interpretation, of the policy space for states to counter supply chain disruption and prepare for the longer round of resilience. Though seeking to reintroduce the “aims and effects” back to the non-discriminatory jurisdiction seems improbably,²⁷³ sneaking in some of the consideration through alternative comparator²⁷⁴ or “similarly-situated”²⁷⁵ theories would not be a bad approach. However, this does not mean for a Pandora’s box for ever expanding national security or exceptionalism, it is still pivotal to draw a line between legitimate risk-based governance and opportunistic discrimination. Several inquiries can be asked: whether the measure targets a demonstrable vulnerability rather than a generalized desire to privilege domestic or certain foreign production; whether it is calibrated to reduce dependency without imposing unnecessary exclusion based on political grounds; whether the state could plausibly pursue comparable resilience gains through less discriminatory alternatives, such as horizontal investment in infrastructure, stockpiling, diversification incentives keyed to concentration rather than origin, or cooperative crisis-response mechanisms; and whether the measure is time-bound or subject to review so that emergency rationales do not harden into permanent hierarchy? Like the current extensive domestic supply chain industry policy has shown, they are operable and generate no less effects than confining to repulsive discriminatory conditionalities.

C. The Way Forward

1. Domestic Self-Discipline

International law is most effective when it is consolidated into domestic law and implemented through enforceable domestic procedures.²⁷⁶ Domestication can proceed in two ways. One approach operates at the level of individual resilience measures. It calls for

²⁷³ See the blatant rejection of the test in Appellate Body Report, *European Communities—Measures Affecting Asbestos and Asbestos-Containing Products*, ¶¶ 100-103, WTO Doc. WT/DS135/AB/R (adopted Mar. 12, 2001).

²⁷⁴ Henrik Horn & Joseph H.H. Weiler, *EC—Asbestos: European Communities—Measures Affecting Asbestos and Asbestos-Containing Products*, 2 WORLD TRADE REV. (SPECIAL ISSUE S1: WTO CASE LAW OF 2001) 14, 25-27 (2003).

²⁷⁵ Julia Ya Qin, *Defining Nondiscrimination under the Law of the World Trade Organization*, 23 BOSTON U. INT’L L.J. 215, 246-51 (2005).

²⁷⁶ Yuji Iwasawa, *Various Means of Enforcement in International Law*, 65 HARV. INT’L L. J. 1, 4 (2023).

restraint in invoking national emergency proclamations, careful policy choices, and a concise, risk-based subsidy design that incorporates comparative advantage and aims for WTO consistency. The JCIB amendment illustrates how targeted policy design can narrow discrimination risks while preserving resilience objectives. A second approach operates at the system level through general subsidy rules that parallel, and sometimes exceed, WTO disciplines. The EU state aid regime reflects this second model. It constrains subsidization through an internal market non-discrimination logic by limiting selective aid that distorts competition and affects trade within the Union, subjecting to centralized Commission oversight and enforcement.²⁷⁷ The United Kingdom's Subsidy Control Act 2022 similarly embeds a non-discrimination constraint by categorically prohibiting subsidies contingent on the use of domestic over imported goods or services, subject to a narrow audiovisual exception.²⁷⁸

It is intriguing to highlight a recent Chinese law practices in subsidies regulation. Under Chinese law, the regulation of subsidies provided by central and local government derives from the anti-unfair competition and anti-monopoly regime. In 1993, "market economy" was incorporated into the Chinese Constitution, signifying that "any organization or individual is prohibited by law from disrupting the social and economic order." Deriving from this principle, the 1993 version of the *Anti-Unfair Competition Law* further stipulates that "the government and its subordinate departments shall not abuse their administrative power" in order to "encourage and protect fair competition."²⁷⁹ The *Anti-Monopoly Law* first enacted in 2007 more systematically stipulates that "administrative power shall not be abused to exclude or restrict competition", and Chapter 5 of the *Anti-Monopoly Law* is devoted to specifying the types of behaviors prohibited from abusing administrative power to exclude or restrict competition.²⁸⁰ However, the indeterminacy of legal responsibility surrounding the abuse of administrative power drew criticism.

In light of the malfunction of the system, the State Council issued the *Opinions on the Establishment of a Fair Competition Review System in the Construction of the Market System* in 2016, seeking to shift from *ex post facto* sanctions at the legal system level to *ex*

²⁷⁷ Consolidated Version of the Treaty on the Functioning of the European Union arts. 107(1), 108(1)-(3), Oct. 26, 2012, 2012 O.J. (C 326) 91-92.

²⁷⁸ Subsidy Control Act 2022, c. 23, § 17(1)-(2) (UK).

²⁷⁹ 金善明 [Shanming Jin], 公平竞争审查机制的制度检讨及路径优化 [Institutional Review and Path Refinement of the Fair Competition Review System], 法学 [L. SCI.] 3, 4 (2019).

²⁸⁰ Fanlongduan Fa (反垄断法) [Anti-Monopoly Law] chapter 5 (adopted by the Standing Comm. Nat'l People's Cong., Aug. 30, 2007, amended June 24, 2022, effective Aug. 1, 2022) 2022 Standing Comm. Nat'l People's Cong. Gazette 623 (China).

ante prevention in the administrative system.²⁸¹ The *Opinions* required the State Council to review the “regulations, normative documents and other policy measures involving the economic activities of market players” as well as the drafting process of the “administrative regulations and other policies and measures formulated by the State Council, as well as local laws and regulations.”²⁸²

The Fair Competition Review System established under the 2016 *Opinions* was later legislated into the *Anti-Monopoly Law* in 2022, and the State Council promulgated the *Regulations on Fair Competition Review* in 2024. The regulations set out principles on the fair competition review for laws, administrative regulations, local laws and regulations, rules, other documents, and policy measures promulgated by administrative organs and organizations charged with the function of managing public affairs.²⁸³ Discriminatory practice in subsidization are disciplined from two perspectives:²⁸⁴ prohibition of setting discriminatory charges, fees, prices or subsidies to non-local or imported goods that restricts the free movement of goods and free flow of production factors;²⁸⁵ prohibition of selective and differentiated tax credits, financial incentives or subsidies, and favorable administrative fees, government funds and social insurance premiums to specific operators that affect the cost of production and operation without explicit authorization from laws and administrative regulations or without the approval of the State Council.²⁸⁶

The State Administration for Market Regulation further elaborated on the fair competition review process, particularly providing a negative list for the selective and differentiated tax credits.²⁸⁷ First, three types of prohibitive discriminatory tax credits are

²⁸¹ Guowuyuan Guanyu Zai Shichang Tixi Jianshe Zhong Jianli Gongping Jingzheng Shenchu Zhidu de Yijian (国务院关于在市场体系建设中建立公平竞争审查制度的意见) [Opinions of the State Council on Establishing a Review System for Fair Competition in the Course of Building the Market System] (promulgated by the St. Council, June 1, 2016), 18 St. Council Gaz., Serial No. 1557, June 30, 2016 (China).

²⁸² *Id.*, sec. 3.1.

²⁸³ Gongping Jingzheng Shenchu Tiaoli (公平竞争审查条例) [Regulations on Fair Competition Review] (adopted at the 32d Exec. Meeting of the St. Council, May 11, 2024; promulgated by the St. Council, June 6, 2024, effective Aug. 1, 2024) 18 St. Council Gaz., Issue No. 18, Serial No. 1845, June 30, 2024 (China).

²⁸⁴ Two pieces of work examines the fair competition review designated for the industrial policy. 孟雁北 [Yanbei Meng], 产业政策公平竞争审查论 [Research on the Fair Competition Review of Industrial Policy], 法学家 [THE JURIST] 118 (2018) (arguing that a proportionality test shall be followed in the fair competition review); 丁茂中 [Maozhong Ding], 产业政策的竞争评估研究 [Study on Competitive Evaluation of Industrial Policy], 法学杂志 [L. SCI. MAG.] 70 (2016) (arguing that the fair competition review on industrial policies shall be both qualitative and quantitative).

²⁸⁵ Regulations of the Fair Competition Review art. 9(2).

²⁸⁶ *Id.*, art. 10.

²⁸⁷ Gongping Jingzheng Shenchu Tiaoli Shishi Banfa (公平竞争审查条例实施办法)

elaborated and prohibited: reducing or exempting a particular operator from the obligation to pay taxes; supporting, in disguise, a specific operator's underpayment or non-payment of taxes by unlawfully converting the operator's form of organization; and supporting, in disguise, a specific operator's underpayment or non-payment of taxes by applying approved levies to a specific industrial park.²⁸⁸ Second, the detailed rules also specified that discriminatory financial incentives or subsidies including those directly determining the beneficiary operators or setting up a directory or enterprise database with unclear or unreasonable entry conditions, those based on the operator's form of ownership, form of organization, etc., and those based on the conditions that operators from outside the region move their place of registration to the local region, pay taxes in the local region, and be included in the local statistics, etc.²⁸⁹ Third, governmental organs or local authorities shall not grant preferential policies in factor acquisition by directly identifying the beneficiary operators, or setting up no objective and clear conditions; reduce, defer, or suspend the levying of administrative and institutional fees and governmental funds; reduce, exempt, or suspend the levying of social insurance fees.²⁹⁰

The development on disciplining domestic subsidies at least shows a positive effort of China in reducing distortive and inefficient subsidy regimes, parallel to other domestic constitutional restriction for discrimination of trade within a state. Another positive trajectory of the regime is that it moves to comply with international rules: the *Fair Competition Review* effectively restricted the discrimination between local/non-local, among different non-locals, between local/imported, and among different imports. However, the system might not be as effective as it reveals. The "self-scrutinization" procedure render this mechanism logically and procedurally circular since the government will only review its own subsidy regime at its own discretion, coupling with lack of transparency and public participation.²⁹¹ Additionally, a significant portion of subsidies

[Implementation Measures for the Regulations on Fair Competition Review] (promulgated by the State Admin. for Mkt. Reg., Feb. 28, 2025, effective Apr. 20, 2025) 12 St. Council Gaz., Serial No. 1875, Apr. 30, 2025 (China).

²⁸⁸ *Id.*, art. 18.

²⁸⁹ *Id.*, art. 19.

²⁹⁰ *Id.*, art. 20.

²⁹¹ 刘继峰 [Jifeng Liu], 论公平竞争审查制度中的问题与解决 [On Problems and Solutions of the Fair Competition Review System], 价格理论与实践 [PRICE: THEORY & PRAC.] 31, 32 (2016); 李俊峰 [Junfeng Li], 公平竞争自我审查的困局及其破解 [The Dilemma and Solutions to the Fair Competition Self-Review System], 华东政法大学学报 [ECUPL J.] 118, 120 (2017); 殷继国 [Jiguo Yin], 我国公平竞争审查模式的反思及其重构 [Reflection and Restructure of the Models of Fair Competition Review System], 政治与法律 [POL. SCI. & L.] 123, 124 (2020).

would be provided through loans and transactions with the Chinese state-owned banks or enterprises as reasoned by the Appellate Body's compliance report in *U.S.–CVD*.²⁹² The Fair Competition Review does not touch upon whether subsidies provided by these state-owned subjects are subject equally the requirement of non-discrimination when they are choosing the provider of raw materials or whichever project to fund.

2. *Responses Through Unilateral Countervailing Measures*

States can address discriminatory supply chain resilience through unilateral countervailing measures. These measures serve as instruments of defensive trade policy, enabling affected countries to offset the adverse effects of subsidies that distort trade and discriminate based on the origin of products or producers. Within the WTO framework, the imposition of countervailing duties requires the existence of a “countervailable subsidy,” injury to a member’s interests, and a causal link between the subsidy and the injury.

It is pivotal to first point out that discriminatory subsidies might already be covered by Article 5(b) of the SCM Agreement. Relying on either the “violation” or “non-violation” nullification or impairment argument, WTO members could already impose countervailing measures on the actionable discriminatory subsidies, so long as systematic offsetting/counteracting of the tariff concession is caused by the subsidy measure.²⁹³ Second, following our analysis of supply chain resilience subsidies with regards to their discriminatory impacts, different procedures and evidential requirements shall be stipulated. For subsidies contingent upon using certain domestic and foreign inputs, since their legal rationale is comparatively straightforward, countervailing measures shall be permitted if a state adopts such subsidies that are already prohibited under the WTO framework. For

²⁹² Appellate Body Report, *United States–Countervailing Duty Measures on Certain Products from China*, ¶¶ 7.46-52, DSU Article 21.5 Proceeding, WTO Doc. WT/DS437/AB/RW (adopted Aug. 15, 2019).

²⁹³ The doctrine was reflected in the *U.S.–Offset Act* case. In the said case, the Panel referring to the pre-WTO case of *EEC–Oilseeds*, the only case concerning a subsidy challenged under the “non-violation” nullification or impairment, and endorsed the *EEC–Oilseeds* test that countries “must ... be assumed to base their tariff negotiations on the expectation that the price effect of the tariff concessions will not be systematically offset” and that it hinged upon the Panel to determine whether “it was reasonable for the United States to expect that the Community would not introduce subsidy schemes systematically counteracting the price effect of the tariff concessions.” Mexico’s argument failed since “the CDSOA does not provide for product-specific subsidies ... and because the amount of such subsidies is not directly linked to the level of tariff concession made for the specific product at issue.” However, for discriminatory subsidies within our discussion, they are product-specific and the impact on tariff concession could be ascertained. Panel Report, *United States–Offset Act*, ¶¶ 7.127-28, WTO Doc. WT/DS217/R, WT/DS234/R (circulated Sep. 16, 2002) (quoting *EEC–Oilseeds* Panel Report, ¶ 147).

subsidies discriminately benefiting producers, cautions shall be maintained. First, where subsidies benefit consumers and thereby indirectly confer a competitive advantage on producers, the competitive relationship test under Articles I and III of the GATT governs. If such a measure distorts the conditions of competition between like products of different origin, it violates the MFN or national treatment obligations. In these cases, countervailing measures should be authorized without the need to prove material injury. The legal rationale lies in restoring equality of competitive opportunities rather than compensating for loss. Second, where subsidies are granted directly to producers, the appropriate legal examination is the serious prejudice test under Article 6.3 of the SCM Agreement. Here, countervailing measures should be permitted only upon evidence that the subsidy displaces or impedes imports of like products from other members. This test reflects the higher threshold associated with tracing how producer-level financial contributions translate into market distortion. As WTO jurisprudence shows, this requires demonstrable impact on trade flows or market shares, and a discernible causal nexus between the subsidy and its adverse effects.

3. *Development of International Rules*

Regulating subsidies is one of the common features of bilateral and regional free trade agreements (“FTAs”). Research by Luca Rubini examined the 283 FTAs signed between 1957 and early February 2016 and found that nearly all FTAs include subsidy provisions (269 out of 283).²⁹⁴ Though the prevalence of the subsidy regulation is observable, Rubini concluded that “in most cases, PTAs simply refer to WTO subsidy disciplines. Additional commitments are largely of an ancillary or procedural – not substantive – nature.”²⁹⁵ The two extremes are the European Union-type FTAs and the U.S.-type FTAs. For the former, the most advanced subsidy and state aid disciplines are explored, while for the latter, a minimal approach to subsidy regulation was adopted.²⁹⁶

The European Union-type FTAs include stricter, more targeted, and enforceable rules, particularly in response to growing concerns over industrial overcapacity and non-market economies. For instance, the EU-Japan Economic Partnership Agreement (“EPA”) builds upon the WTO framework by requiring transparency and regular notification of subsidies that affect trade and investment.²⁹⁷ The agreement obliges both parties to disclose

²⁹⁴ Rubini, *supra* note 118, at 440.

²⁹⁵ *Id.*, 458.

²⁹⁶ *Id.*

²⁹⁷ A general discussion on the subsidy chapter of the EU-Japan EPA, see Aya Iino, *Disciplining Subsidies Through Free Trade Agreements (FTAs): Emerging Developments*

information on the legal basis, form, and amount of subsidies upon request, thus enhancing enforceability beyond the WTO's largely aspirational notification regime. Substantively, the EU-Japan EPA introduces two types of prohibited subsidies in addition to the WTO framework: a guarantee of debts or liabilities of an enterprise, without any limitation as to the amount and duration of such guarantee; subsidies for restructuring an ailing or insolvent enterprise without the enterprise having prepared a credible restructuring plan.²⁹⁸ These two types of subsidies will be prohibited if they "have or could have a significant negative effect on trade or investment between the Parties."²⁹⁹ Similar prohibition was found in the EU-South Korea FTA,³⁰⁰ EU-Singapore FTA,³⁰¹ and EU-Vietnam FTA,³⁰² with modifications on the onus and burden of proof on the "negative effect on trade or investment."

Similar to the two types of "WTO+" prohibited subsidies, it is possible to introduce another new type of prohibition on discriminatory subsidies within the regional and bilateral agreements. As has been shown by the recent practice of fishery subsidies,³⁰³ the interactions between subsidies discipline in FTAs and the WTO rules are potentially beneficial, and lessons at regional settings could "help shape international trade law to be more compatible with the changing international economic order."³⁰⁴ Among the three types of discriminatory subsidies currently not prohibited under the SCM Agreement, we argue that subsidies contingent upon using certain designated foreign inputs shall be explicitly prohibited. Such subsidies confer advantages on products originating from one country that are not extended immediately and unconditionally to like products from other countries, and alter the competitive relationship of the relevant markets by discriminatorily awarding those final product producers in selecting certain foreign products. In so doing, the reasoning underpinning Article 3.1(b) of the SCM Agreement shall be analogized to prohibit its distortive effects.

in Japan's FTAs and Their Implications, 24 Ger. L.J. 179 (2023).

²⁹⁸ Agreement Between the European Union and Japan for an Economic Partnership, 2022 O.J. (L 330) 3, art. 12.7.

²⁹⁹ *Id.*

³⁰⁰ Free Trade Agreement Between the European Union and Its Member States, of the One Part, and the Republic of Korea, of the Other Part, 2011 O.J. (L 127) 6, art. 11.11.

³⁰¹ Free Trade Agreement Between the European Union and the Republic of Singapore, 2019 O.J. (L 294) 3, art. 11.7

³⁰² Free Trade Agreement Between the European Union and the Socialist Republic of Viet Nam, 2020 O.J. (L 186) 3, art. 10.9(1).

³⁰³ Chen-Ju Chen, *Current Developments of Fisheries Subsidies Regulations After the Failure of WTO Doha Round Negotiations with a Focus on the Asia-Pacific Region*, 17 ASIAN J. WTO & INT'L HEALTH L. & POL'Y 41 (2022).

³⁰⁴ Iino, *supra* note 297, at 180.

However, we need to carefully address Rubini's finding that "basic statistics show that general national treatment provisions have been increasingly seen as not appropriate for regulating subsidies."³⁰⁵ Rubini distinguished between FTAs that devise their own national treatment clause and those that incorporate Article III of the GATT. He calculated that the absolute number of FTA-specific national treatment clause fluctuated throughout the research period (38 in the pre-1995 period, 26 in the 1995-1999 and 2000-2004 periods, 31 in 2005-2009, and 19 in 2010-2017), while the absolute number of FTA-specific national treatment clause for goods gradually decreased (36 in pre-1995, 25 in 1995-1999, 15 in 2000-2004, 16 in 2005-2009, and 5 in 2010-2017).³⁰⁶ Rubini's research excluded the latter practice of Article III incorporation "because of its sweeping exclusion of subsidies to domestic producers (paragraph 8(b))."³⁰⁷ He reasoned that "parties to PTAs shelter subsidies from national treatment obligations largely by incorporating GATT Article III (with its built-in exclusion of domestic subsidies)."³⁰⁸ However, this analysis in technical-legal terms is not fully persuasive in light of the jurisprudence and drafting history of Article III:8(b) of the GATT. As we have shown, the politically driven and potentially "honest mistake" of the domestic producer exception shall be narrowly interpreted and only exonerates subsidies through the form of paying expenditure of revenue by a government to designated final product producers. While virtually all FTAs include a national treatment clause (either individualized or incorporated), regulating discrimination in subsidization contrarily has a solid foundation in FTAs.

VI. CONCLUSION

Supply chain resilience has shifted from a managerial aspiration to a central theme of industrial policy. This paper has argued that subsidies are a particularly attractive resilience tool since they can stimulate investment in capacity, redundancy, information, and rapid adjustment. These are areas where markets may undersupply because resilience benefits spill across firms and tiers. Yet the same instrument can cause inefficiency by encouraging redundancy without real security gains and amplify rent-seeking. Supply chain resilience policies could also be harmful to domestic and international welfare if applied wrongly.

Doctrinally, the WTO system remains the baseline discipline for industrial policy. The SCM Agreement leaves meaningful space for industrial policy since most resilience subsidies are not prohibited *per se* and they are typically contested through demanding

³⁰⁵ Rubini, *supra* note 118, at 442.

³⁰⁶ *Id.*, at 442-443.

³⁰⁷ *Id.*, at 442, note 19.

³⁰⁸ *Id.*, at 443.

adverse-effects analysis. At the same time, the GATT's non-discrimination obligations constrain the most direct forms of discriminatory conditionality. This is especially pertinent and affects market conditions where subsidy design steers sourcing away from non-designated imports. At the same time, the analysis clarifies that reliance on the GATT exceptions as a general safety valve is often fragile, and that narrow, commodity-specific commodity agreements rarely supply a stable justification for MFN departures.

Normatively, the Article defends the continuing value of non-discrimination as a device that stabilizes bargains, disciplines rent-seeking selectivity, and preserves the welfare gains of diversified production networks. The policy task is therefore to reconceptualize permissible resilience subsidies through workable constraints: domestic self-discipline that cabins discriminatory conditionality, calibrated unilateral responses where injury is real and demonstrable, and incremental development of international rules that recognize resilience objectives and state's respective comparative advantage without normalizing permanent, partner-only favoritism.